

No-Regret Learning in Stackelberg Security Games with an Unknown, Bounded Set of Attacker Types

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1 Abstract

Stackelberg Security Games (SSGs) model the strategic allocation of limited defender resources against attackers whose preferences are often uncertain. Balcan et al. [2] designed no-regret online learning algorithms for the *repeated* variant of SSGs against an adversarial sequence of attackers drawn from a finite, *known* set K , and proved a matching linear-regret lower bound when K is unbounded. A natural real-world middle ground is left open: the defender does not know K a priori, but has a bound $K_{\max}$ on its size. We study this setting in the full-information feedback model. We propose GROW-CATALOGUE, an algorithm that incrementally learns new attacker types as they arrive, re-partitions the coverage polytope, and runs an anytime multiplicative-weights sub-routine on the resulting extreme-point set. Our central technical contribution is a full proof that GROW-CATALOGUE achieves expected regret

$$O\left(\sqrt{T \cdot K_{\max} \cdot (n^2 + K_{\max} \log n)}\right)$$

against any oblivious adversarial sequence of length T . The rate matches that of Balcan et al. [2] with k replaced by $K_{\max}$ and is sublinear whenever $K_{\max} \cdot (n^2 + K_{\max} \log n) = o(T)$. Experiments on synthetic SSG instances confirm the predicted $\sqrt{T}$ scaling, and the empirical regret stays comfortably below the analytic $C\sqrt{T}$ envelope. Our implementation and all experiment scripts are at github.com/aditisingh1053/Security_Games.

2 Introduction

In a Stackelberg Security Game (SSG), a defender allocates a scarce set of resources to protect n potential targets against a strategic attacker who, observing the defender’s randomized deployment, chooses the best target to attack [20]. Deployed systems built around this model include the US Coast Guard’s port protection patrols, the Federal Air Marshals assignment system, the LAX airport patrol schedule, and anti-poaching patrols in several African national parks [20, 21]. In every one of these applications the defender must commit to its deployment before the attack happens, and the attacker’s payoff function encodes its preferences over targets.

The *classical* SSG analysis assumes that the attacker’s payoff function is fully known to the defender. The optimal commitment can then be computed by a linear program [e.g., 19]. In practice, however, the defender faces considerable uncertainty about the attacker. There may be several distinct threat actors (e.g., different poacher groups, different types of smugglers), and their utilities are only imperfectly known. A principled response, pioneered by Balcan et al. [2], is to drop the single-type assumption entirely and instead ask for

a *no-regret* online algorithm against an adversarially chosen sequence of attacker types, drawn from a known finite set K with $|K| = k$. Under full-information feedback, they design an algorithm whose expected cumulative regret is bounded by $O(\sqrt{Tn^2k \log(nk)})$, which is sublinear in the horizon T , and they prove a matching linear-regret lower bound when $|K|$ is unrestricted [2, Theorem 7.1].

The assumption that the set K itself is known to the defender in advance is, however, a strong one. In realistic deployments the defender typically does *not* have a complete list of attacker types at the start. What it may have is an *upper bound* $K_{\max}$ on the number of distinct types it expects to encounter (for instance, from threat intelligence or from physical constraints of the domain). The question that motivates this project is therefore:

Can the defender achieve sublinear regret when K is unknown a priori, subject to $|K| \leq K_{\max}$?

This question lies squarely in the gap left open between the *known- K* upper bound and the *unrestricted- K* lower bound of Balcan et al. [2]. It is not answered by any of the existing work, and resolving it is the main goal of the present report.

2.1 Contributions

Our contributions are the following.

- **Problem formalization (Section 4).** We formalize the unknown- K variant of the repeated Stackelberg security game, identifying the minimal modification of the model of Balcan et al. [2] that captures it.
- **Algorithm (Section 5).** We propose GROW-CATALOGUE, an online algorithm that at each round maintains a *catalogue* $C_t \subseteq K$ of attacker types observed so far, recomputes the set $\mathcal{E}(C_t; \epsilon)$ of ϵ -approximate extreme points of the partition of $\mathcal{P}$ induced by C_t , and runs an anytime multiplicative-weights sub-routine over $\mathcal{E}(C_t)$. When a new type is observed, the catalogue is updated and the sub-routine is restarted on the enlarged expert set.
- **Main theorem and a complete, detailed proof (Sections 5.2 and 6).** The technical core of this report is a self-contained proof that GROW-CATALOGUE achieves expected regret at most $O(\sqrt{T \cdot K_{\max} \cdot (n^2 + K_{\max} \log n)})$ against any oblivious adversarial sequence of length T under full-information feedback. The rate matches Theorem 5.1 of Balcan et al. [2] with k replaced by $K_{\max}$; for constant $K_{\max}$ it is $\tilde{O}(n\sqrt{T})$.
- **Regime analysis (Corollaries 7 and 8).** We characterize the parameter regime in which the bound is meaningful ($K_{\max} \cdot (n^2 + K_{\max} \log n) = o(T)$), and we prove an instance-adaptive refinement in which the leading factor depends on

the number K' of types *actually* observed rather than on $K_{\max}$.

- **Experiments (Section 7).** We implement GROW-CATALOGUE and two reference baselines, and evaluate them on synthetic SSG instances. The empirical regret scales as predicted by the theorem, staying comfortably below the analytic $C\sqrt{T}$ envelope.

2.2 Related Work

Classical SSGs and attacker uncertainty. The optimal commitment for a known attacker can be computed by an LP [6]. Uncertainty about the attacker has been handled by *robust optimization*, which plans against worst-case payoffs [19], and by *Bayesian SSGs*, which require a prior over attacker types [13, 16]. Both approaches rest on priors or constraints that are often hard to validate. Letchford et al. [14] give a query-efficient learning algorithm for a *single* fixed but unknown type, and Blum et al. [3] extend this to a setting that adapts round-by-round to one fixed type.

Online learning against adversarial attacker sequences. Balcan et al. [2] pivot from the single-type paradigm by allowing the sequence of attackers to be adversarially chosen from a known finite set K . For full information they give a polynomial-weights algorithm with regret $O(\sqrt{Tn^2k \log(nk)})$ that partitions the coverage polytope into best-response regions and runs experts over the ϵ -approximate extreme points; for bandit feedback they use a barycentric spanner construction to give an $O(T^{2/3})$ bound. Crucially, they also prove a linear lower bound when $|K|$ is unrestricted, which forces any non-trivial extension to restrict the attacker type set somehow. We take the weakest restriction: a known cardinality bound $K_{\max}$.

Online learning with changing expert sets. The subroutine inside GROW-CATALOGUE is an anytime variant of Polynomial Weights / Hedge [4, 7, 15]. We restart this subroutine each time the catalogue grows; simpler algorithms for tracking a growing set of experts exist [17], and for tracking a shrinking set [8], but both would only shave lower-order logarithmic factors.

Other recent directions. Haghtalab et al. [9] analyze no-regret learning against strategic (non-myopic) attackers in a single-type query model. Harris et al. [10] extend to a contextual Stackelberg setting in which both players observe side information. Neither addresses the *unknown K with bounded cardinality* question that we pose here.

3 Model

We adopt the repeated-SSG notation of Balcan et al. [2, Sec. 2]; our notation is compatible with theirs.

3.1 Game primitives

The game has a finite horizon $T \in \mathbb{N}$ and n targets $N = \{1, \dots, n\}$. The defender has R resources and a collection $\mathcal{D} \subseteq 2^N$ of *schedules*, where each schedule is a subset of

targets that a single resource can defend simultaneously. An *assignment function* $A : R \rightarrow 2^{\mathcal{D}}$ records the schedules that each resource may be assigned to. A *pure strategy* is any valid resource-to-schedule assignment; let m denote the number of pure strategies. A *mixed strategy* is a distribution $\mathbf{s} \in \Delta_m$ over pure strategies, and its induced *coverage probability vector* is $\mathbf{p} = M\mathbf{s} \in [0, 1]^n$, where $M \in \{0, 1\}^{n \times m}$ has $M_{ij} = 1$ iff the j -th pure strategy covers target i . Write $\mathcal{P}$ for the set of valid coverage probability vectors; Every coverage vector $\mathbf{p}$ is realizable by some mixed strategy, and conversely every mixed strategy induces a unique $\mathbf{p} \in \mathcal{P}$.

Defender payoffs. For each target i , the defender has a payoff $u_d^c(i) \in [-1, 1]$ if i is attacked *and* covered, and a payoff $u_d^u(i) \in [-1, 1]$ if i is attacked *and* uncovered. The defender's expected utility when target i is attacked under coverage $\mathbf{p}$ is therefore

$$U_d(i, \mathbf{p}) = u_d^c(i) p_i + u_d^u(i) (1 - p_i), \quad (1)$$

which is *linear* in $\mathbf{p}$ for each fixed target i .

3.2 Attacker types and best response

An attacker *type* α is a pair $(u_\alpha^c, u_\alpha^u) \in [-1, 1]^n \times [-1, 1]^n$ of per-target utilities following the same “covered/uncovered” convention as the defender. Its expected utility for attacking target i under coverage $\mathbf{p}$ is $U_\alpha(i, \mathbf{p}) = u_\alpha^c(i) p_i + u_\alpha^u(i) (1 - p_i)$. Observing $\mathbf{p}$, type α *best-responds*:

$$b_\alpha(\mathbf{p}) = \arg \max_{i \in N} U_\alpha(i, \mathbf{p}), \quad (2)$$

with ties broken arbitrarily but consistently. Note that b_α is piecewise constant in $\mathbf{p}$, which makes the defender's realized payoff $U_d(b_\alpha(\mathbf{p}), \mathbf{p})$ a discontinuous function of $\mathbf{p}$. This non-continuity is the central technical difficulty that distinguishes SSGs from vanilla online linear optimization, and it is what drives the polytope-based machinery of Balcan et al. [2].

3.3 Repeated game and regret

There is a (possibly unknown) set K of attacker types. An *oblivious* adversary chooses a sequence $\mathbf{a} = a_1, \dots, a_T \in K^T$ before the game starts; the sequence is fixed but unknown to the defender. At each round $t \in \{1, \dots, T\}$ the defender commits $\mathbf{p}_t \in \mathcal{P}$, attacker a_t plays $b_{a_t}(\mathbf{p}_t)$, and the defender receives the random payoff $U_d(b_{a_t}(\mathbf{p}_t), \mathbf{p}_t)$. We assume *full-information feedback*: at the end of round t the defender observes the identity of a_t , and the first time any type α is observed, its utility functions are revealed as well.

The defender's benchmark is the best fixed coverage vector in hindsight:

$$\mathbf{p}^* = \arg \max_{\mathbf{p} \in \mathcal{P}} \sum_{t=1}^T U_d(b_{a_t}(\mathbf{p}), \mathbf{p}). \quad (3)$$

The expected regret of an online algorithm that plays $\mathbf{p}_t$ at round t is

$$\text{Regret}(T) = \sum_{t=1}^T U_d(b_{a_t}(\mathbf{p}^*), \mathbf{p}^*) - \mathbb{E} \left[\sum_{t=1}^T U_d(b_{a_t}(\mathbf{p}_t), \mathbf{p}_t) \right], \quad (4)$$

where the expectation is over the algorithm's internal randomization.

3.4 Preliminaries: polytope partition and $\mathcal{E}(C; \epsilon)$

The following definitions, taken from Balcan et al. [2, Sec. 4], are essential. For any catalogue $C \subseteq K$, a type $\alpha \in C$, and a target $i \in N$, the region in which α 's best response is i is

$$\mathcal{P}_i^\alpha(C) = \{\mathbf{p} \in \mathcal{P} : b_\alpha(\mathbf{p}) = i\}, \quad (5)$$

which is the intersection of $\mathcal{P}$ with the $n-1$ linear inequalities $U_\alpha(i, \mathbf{p}) \geq U_\alpha(j, \mathbf{p})$ for $j \neq i$; hence a convex polytope. For a response profile $\sigma : C \rightarrow N$ we write

$$\mathcal{P}_\sigma(C) = \bigcap_{\alpha \in C} \mathcal{P}_{\sigma(\alpha)}^\alpha(C). \quad (6)$$

The non-empty sets $\mathcal{P}_\sigma(C)$ form a partition of $\mathcal{P}$ into convex regions in which the best response of every type in C is held fixed; within any such region, the defender's payoff under a fixed attacker type is a linear function of $\mathbf{p}$.

We will need three results from Balcan et al. [2], restated below in our notation and cited explicitly at every use. Lemmas 1 and 2 describe the approximate extreme-point set; Proposition 3 is the standard Polynomial-Weights online learning guarantee, adapted to the present loss range.

Lemma 1 (2, Lemma 4.3). *For every $\epsilon > 0$ and every catalogue $C \subseteq K$, there is a finite set $\mathcal{E}(C; \epsilon) \subseteq \mathcal{P}$ of points, called the ϵ -approximate extreme points of the partition, such that for any sequence $\mathbf{a}' = a'_1, \dots, a'_{T'}$ with $a'_t \in C$ for all t ,*

$$\max_{\mathbf{p} \in \mathcal{E}(C; \epsilon)} \sum_{t=1}^{T'} U_d(b_{a'_t}(\mathbf{p}), \mathbf{p}) \geq \max_{\mathbf{p} \in \mathcal{P}} \sum_{t=1}^{T'} U_d(b_{a'_t}(\mathbf{p}), \mathbf{p}) - 2\epsilon T'.$$

Lemma 2 (2, Lemma 4.4). *The set $\mathcal{E}(C; \epsilon)$ of Lemma 1 can be chosen so that $|\mathcal{E}(C; \epsilon)| \in O((2^n + |C|n^2)^n \cdot n^{|C|})$.*

Proposition 3 (2, Proposition 1). *For any fixed expert set $\mathcal{M}$ and any sequence of losses in $[-\kappa, \kappa]$, there is an online learning algorithm (an anytime variant of Polynomial Weights / Hedge; see 4, Sec. 2) whose expected regret after T' rounds is at most $O(\sqrt{T' \kappa \log |\mathcal{M}|})$, without requiring T' to be known in advance.*

4 The Unknown- K Setting

We now state the problem that our algorithm and theorem address.

Problem 4 (Unknown- K repeated SSG). A repeated SSG as in Section 3 is played, with two modifications.

1. The set K of attacker types is not revealed to the defender at the start; however, the defender is given a constant $K_{\max} \geq 1$ with the promise $|K| \leq K_{\max}$.
2. Feedback is full information: after round t the defender observes a_t , and the first time any type α appears, its utility vectors (u_α^c, u_α^u) are revealed.

The benchmark $\mathbf{p}^*$ and regret $\text{Regret}(T)$ are as in Equations (3) and (4).

Algorithm 1 GROW-CATALOGUE (full information).

Require: horizon T , bound $K_{\max}$, game primitives, default $\mathbf{p}_{\text{def}} \in \mathcal{P}$

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1:  $C \leftarrow \emptyset$ ;  $\mathcal{E} \leftarrow \emptyset$ ;  $\epsilon \leftarrow 1/\sqrt{T}$ ;  $t_{\text{loc}} \leftarrow 0$ 
2: for  $t = 1, \dots, T$  do
3:   if  $\mathcal{E} = \emptyset$  then
4:     play  $\mathbf{p}_t \leftarrow \mathbf{p}_{\text{def}}$ 
5:   else
6:     sample  $\mathbf{p}_t \sim q_t$ , where  $q_t$  is the current PW distribution
       over  $\mathcal{E}$ 
7:   end if
8:   observe  $a_t$ 
9:   if  $a_t \notin C$  then ▷ discovery round: new attacker type
10:     $C \leftarrow C \cup \{a_t\}$ ; record  $(u_{a_t}^c, u_{a_t}^u)$ 
11:     $\mathcal{E} \leftarrow \mathcal{E}(C; \epsilon)$  ▷ recompute extreme points
12:    reset PW on  $\mathcal{E}$ ;  $t_{\text{loc}} \leftarrow 0$ 
13:   else ▷ routine round
14:     $t_{\text{loc}} \leftarrow t_{\text{loc}} + 1$ 
15:    compute  $\ell_t(\mathbf{p}) = -U_d(b_{a_t}(\mathbf{p}), \mathbf{p})$  for  $\mathbf{p} \in \mathcal{E}$ 
16:     $\eta \leftarrow \sqrt{\log |\mathcal{E}| / t_{\text{loc}}}$ 
17:     $w_{\mathbf{p}} \leftarrow w_{\mathbf{p}} \cdot e^{-\eta \ell_t(\mathbf{p})}$  for all  $\mathbf{p} \in \mathcal{E}$ 
18:   end if
19: end for
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Remark 5 (Necessity of the bound $K_{\max}$). Balcan et al. [2, Theorem 7.1] exhibit, for every T , a game with $|K| < 2^{T+1}$ on which every online algorithm has expected regret at least $T/2$. Hence some finite bound on $|K|$ is *necessary* for sublinear regret; Problem 4 takes the weakest such bound, namely a known constant $K_{\max}$.

5 Algorithm and Main Result

5.1 The Grow-Catalogue algorithm

We describe GROW-CATALOGUE at a high level and then give the pseudocode in Algorithm 1. At any round t the defender holds a catalogue $C_t \subseteq K$ of all types seen so far. Conceptually, the defender runs a small instance of the *known- K* algorithm of Balcan et al. [2, Sec. 5] on the catalogue C_t , pretending that C_t is the full attacker set. Concretely:

- The algorithm maintains the ϵ -approximate extreme-point set $\mathcal{E}_t = \mathcal{E}(C_t; \epsilon)$ of Lemma 1. We set $\epsilon = 1/\sqrt{T}$; this balances the $2\epsilon T$ approximation term in Lemma 1 against the $\sqrt{T \log N_{\max}}$ regret term of Proposition 3.
- An anytime Polynomial-Weights sub-routine of Proposition 3 runs over $\mathcal{E}_t$, updating via a time-varying learning rate $\eta_{t_{\text{loc}}} = \sqrt{\log |\mathcal{E}_t| / t_{\text{loc}}}$ where t_{loc} is a local counter reset at every catalogue change. This is the standard anytime schedule; see Cesa-Bianchi and Lugosi [4, Sec. 2].
- When a new attacker type α appears, the catalogue grows to $C_t \cup \{\alpha\}$, the approximate extreme-point set is recomputed, and the sub-routine is restarted on the enlarged expert set. The single round of first contact with α contributes a bounded additive amount to the regret that we will absorb.

5.2 Main Result

Theorem 6 (Main). *Let the defender play Algorithm 1 in the unknown- K full-information setting of Problem 4. For any oblivious adversarial sequence $\mathbf{a} \in K^T$ with $|K| \leq K_{\max}$, the*

expected regret satisfies

$$\mathbb{E}[\text{Regret}(T)] \leq O\left(\sqrt{T \cdot K_{\max} \cdot (n^2 + K_{\max} \log n)}\right). \quad (7)$$

In particular, if $K_{\max} = O(1)$ then $\mathbb{E}[\text{Regret}(T)] = \tilde{O}(n\sqrt{T})$, matching the known- K bound of Balcan et al. [2, Theorem 5.1] up to logarithmic factors.

We prove Theorem 6 in detail in Section 6. Two useful corollaries of the proof (the regime of applicability and an instance-adaptive refinement) are stated after the proof in Section 6.1.

6 Proof of Theorem 6

The proof closely tracks the structure of the known- K proof of Balcan et al. [2, Sec. 5] but has two non-trivial deviations: (i) the expert set is no longer static but grows over time, and (ii) the discovery rounds introduce an extra bounded additive term. We account for both the deviations in our proof.

Proof:- Fix an oblivious adversarial sequence $\mathbf{a} = a_1, \dots, a_T \in K^T$ with $|K| \leq K_{\max}$. Let $K' = |\{a_1, \dots, a_T\}| \leq K_{\max}$ denote the number of distinct types that actually appear in $\mathbf{a}$. The proof proceeds in ten steps.

Step 1 (Epoch decomposition). Let $\tau_1 < \tau_2 < \dots < \tau_{K'}$ be the round indices at which each new attacker type is first observed, i.e., for each $i \in \{1, \dots, K'\}$, round τ_i is the smallest t such that $a_t \notin \{a_1, \dots, a_{t-1}\}$. We define the *epochs*

$$\text{Ep}_i = \{\tau_i + 1, \tau_i + 2, \dots, \tau_{i+1} - 1\}, \quad i = 1, \dots, K', \quad (8)$$

with the convention $\tau_{K'+1} := T + 1$, and write $L_i = |\text{Ep}_i|$. Let $\mathcal{D} = \{\tau_1, \dots, \tau_{K'}\}$ be the set of *discovery rounds*. By construction,

$$[1, T] = \mathcal{D} \sqcup \bigcup_{i=1}^{K'} \text{Ep}_i,$$

and $|\mathcal{D}| + \sum_{i=1}^{K'} L_i = T$ with $|\mathcal{D}| = K' \leq K_{\max}$. The epoch decomposition is the critical structural property of the proof: it re-interprets the online problem with a growing expert set as a finite sequence of online problems with *fixed* expert sets, so that the known- K analysis of Balcan et al. [2] can be invoked inside each epoch.

Step 2 (Catalogue is constant inside an epoch). Immediately after round τ_i , the catalogue maintained by Algorithm 1 is $C_i = \{a_{\tau_1}, \dots, a_{\tau_i}\}$, and it is not updated until the next discovery round τ_{i+1} . Hence $C_t = C_i$ for every $t \in \text{Ep}_i$. Moreover, by the defining property of τ_{i+1} , no new type appears in Ep_i , so

$$a_t \in C_i \quad \text{for every } t \in \text{Ep}_i. \quad (9)$$

Denote by $\mathcal{E}_i := \mathcal{E}(C_i; \epsilon)$ the extreme-point set used by the PW sub-routine during Ep_i ; Equation (9) is exactly the hypothesis needed to apply Lemma 1 to C_i and the sub-sequence inside Ep_i .

Step 3 (Within-epoch regret). During Ep_i the algorithm runs a fresh instance of the anytime Polynomial-Weights sub-routine on the fixed expert set $\mathcal{E}_i$ for L_i rounds. The losses are $\ell_t(\mathbf{p}) = -U_d(b_{a_t}(\mathbf{p}), \mathbf{p}) \in [-1, 1]$, so the range parameter is $\kappa = 1$. Proposition 3 therefore yields an absolute constant $c_1 > 0$ such that

$$\begin{aligned} & \max_{\mathbf{p} \in \mathcal{E}_i} \sum_{t \in \text{Ep}_i} U_d(b_{a_t}(\mathbf{p}), \mathbf{p}) \\ & - \mathbb{E} \left[\sum_{t \in \text{Ep}_i} U_d(b_{a_t}(\mathbf{p}_t), \mathbf{p}_t) \right] \leq c_1 \sqrt{L_i \log |\mathcal{E}_i|}. \end{aligned} \quad (10)$$

Step 4 (Within-epoch approximation via Lemma 1).

We now wish to compare the max over $\mathcal{E}_i$ during Ep_i with the *global* benchmark $\mathbf{p}^*$ (the best fixed strategy over the full horizon $[1, T]$). Lemma 1, however, is a statement of the form “ $\max_{\mathbf{p} \in \mathcal{E}(C; \epsilon)}(\cdot) \geq \max_{\mathbf{p} \in \mathcal{P}}(\cdot) - 2\epsilon T'$ ” and, strictly speaking, does *not* directly involve $\mathbf{p}^*$: it compares the max over the finite set $\mathcal{E}(C; \epsilon)$ against the unrestricted max over $\mathcal{P}$, both on the *same* attacker sequence. We therefore proceed in two sub-steps and make the passage to $\mathbf{p}^*$ explicit.

Sub-step 4a (apply Lemma 1 to the epoch sub-sequence). By Equation (9), the sub-sequence $(a_t)_{t \in \text{Ep}_i}$ has length L_i and uses only types in C_i , which is exactly the hypothesis of Lemma 1. Applying the lemma with catalogue C_i and sequence length $T' = L_i$ gives

$$\begin{aligned} & \max_{\mathbf{p} \in \mathcal{E}_i} \sum_{t \in \text{Ep}_i} U_d(b_{a_t}(\mathbf{p}), \mathbf{p}) \\ & \geq \underbrace{\max_{\mathbf{p} \in \mathcal{P}} \sum_{t \in \text{Ep}_i} U_d(b_{a_t}(\mathbf{p}), \mathbf{p})}_{=: \text{OPT}_i} - 2\epsilon L_i. \end{aligned} \quad (11)$$

Here OPT_i is the best cumulative utility achievable on Ep_i by *any* coverage vector in $\mathcal{P}$; crucially, this is the epoch- i optimum, which in general *differs* from the value of $\mathbf{p}^*$. This is the subtle point that distinguishes our setting from the known- K proof of Balcan et al. [2, Sec. 5]: in their analysis, Lemma 1 is applied once on the full horizon, so $\text{OPT} = \sum_{t=1}^T U_d(b_{a_t}(\mathbf{p}^*), \mathbf{p}^*)$ by definition of $\mathbf{p}^*$ and no separate passage is needed. In our per-epoch application, OPT_i is an epoch-local quantity, potentially larger than $\mathbf{p}^*$'s value on the epoch.

Sub-step 4b (pass from OPT_i to $\mathbf{p}^$).* Since $\mathbf{p}^* \in \mathcal{P}$, the coverage vector $\mathbf{p}^*$ is a *feasible* competitor in the maximization defining OPT_i . Therefore

$$\text{OPT}_i \geq \sum_{t \in \text{Ep}_i} U_d(b_{a_t}(\mathbf{p}^*), \mathbf{p}^*). \quad (12)$$

Inequality Equation (12) is *not* a consequence of Lemma 1 but of the trivial fact that the global benchmark $\mathbf{p}^*$ is an element of the set over which the epoch-local optimum is taken. Chaining Equation (11) with Equation (12),

$$\begin{aligned} & \max_{\mathbf{p} \in \mathcal{E}_i} \sum_{t \in \text{Ep}_i} U_d(b_{a_t}(\mathbf{p}), \mathbf{p}) \\ & \geq \sum_{t \in \text{Ep}_i} U_d(b_{a_t}(\mathbf{p}^*), \mathbf{p}^*) - 2\epsilon L_i. \end{aligned} \quad (13)$$

Step 5 (Per-epoch combined bound). Chaining Equations (10) and (13), for every epoch $i \in \{1, \dots, K'\}$ we obtain

$$\begin{aligned} & \mathbb{E} \left[\sum_{t \in \text{Ep}_i} U_d(b_{a_t}(\mathbf{p}_t), \mathbf{p}_t) \right] \\ & \geq \sum_{t \in \text{Ep}_i} U_d(b_{a_t}(\mathbf{p}^*), \mathbf{p}^*) - 2\epsilon L_i - c_1 \sqrt{L_i \log |\mathcal{E}_i|}. \end{aligned} \quad (14)$$

Equation (14) is the per-epoch regret bound *against the global benchmark* $\mathbf{p}^*$, and it is the building block for the rest of the proof. Note that despite the fact that $\mathbf{p}^*$ is the benchmark over all T rounds (not just epoch i), we are able to bound our per-epoch regret against $\mathbf{p}^*$ because $\mathbf{p}^* \in \mathcal{P}$ and we applied Lemma 1 on the sub-sequence.

Step 6 (Discovery rounds). For each discovery round $t \in \mathcal{D}$, the algorithm either plays $\mathbf{p}_{\text{def}}$ (if $t = \tau_1$) or samples $\mathbf{p}_t$ from the PW distribution inherited from Ep_{i-1} (if $t = \tau_i$ for $i \geq 2$). In both cases the sampled $\mathbf{p}_t$ is not tuned to the newly-appearing type $a_t = a_{\tau_i}$. However, because the payoff U_d is bounded in $[-1, 1]$, for every $\mathbf{p} \in \mathcal{P}$ and every type α , $U_d(b_\alpha(\mathbf{p}), \mathbf{p}) \in [-1, 1]$, so

$$|U_d(b_{a_t}(\mathbf{p}^*), \mathbf{p}^*) - U_d(b_{a_t}(\mathbf{p}_t), \mathbf{p}_t)| \leq 2. \quad (15)$$

Summing Equation (15) over the at most $K' \leq K_{\max}$ discovery rounds yields

$$\sum_{t \in \mathcal{D}} U_d(b_{a_t}(\mathbf{p}^*), \mathbf{p}^*) - \mathbb{E} \left[\sum_{t \in \mathcal{D}} U_d(b_{a_t}(\mathbf{p}_t), \mathbf{p}_t) \right] \leq 2K_{\max}. \quad (16)$$

This is the only source of additive dependence on $K_{\max}$ that does not come from the main $\sqrt{TK_{\max}}$ term. It is the “price of not knowing K ” at a discovery round.

Step 7 (Uniform bound on $|\mathcal{E}_i|$). By Lemma 2 applied with $|C_i| \leq K_{\max}$,

$$|\mathcal{E}_i| \leq c_2 (2^n + K_{\max} n^2)^n \cdot n^{K_{\max}} =: N_{\max},$$

for an absolute constant c_2 . Taking logarithms,

$$\begin{aligned} \log N_{\max} &= n \log(2^n + K_{\max} n^2) + K_{\max} \log n + O(1) \\ &\leq n^2 + n \log\left(1 + \frac{K_{\max} n^2}{2^{n-1}}\right) + K_{\max} \log n \\ &= O(n^2 + K_{\max} \log n), \end{aligned} \quad (17)$$

where the middle step uses $\log(2^n + K_{\max} n^2) \leq n + \log(1 + K_{\max} n^2 / 2^{n-1})$ and $\log(1 + x) \leq x$ for $x > 0$. Crucially, Equation (17) is uniform in i : $\log |\mathcal{E}_i| \leq \log N_{\max}$ for every epoch. This lets us pull the $\log |\mathcal{E}_i|$ factor in Step 5 outside the sum over epochs.

Step 8 (Summing across epochs). Sum Equation (14) over $i = 1, \dots, K'$ and add Equation (16). The algorithm’s total expected utility is then at least the benchmark’s total utility, minus the discovery term $2K_{\max}$, minus $2\epsilon T$ from the telescoping Lemma-4.3 approximation errors (using $\sum_i L_i \leq T$), minus the sum of the per-epoch PW regret terms. Rearranging,

$$\mathbb{E}[\text{Regret}(T)] \leq 2K_{\max} + 2\epsilon T + c_1 \sqrt{\log N_{\max}} \sum_{i=1}^{K'} \sqrt{L_i}. \quad (18)$$

Step 9 (Cauchy–Schwarz). The last term involves the sum of K' square roots. By the Cauchy–Schwarz inequality applied to $(1, \dots, 1) \in \mathbb{R}^{K'}$ and $(\sqrt{L_1}, \dots, \sqrt{L_{K'}})$,

$$\sum_{i=1}^{K'} \sqrt{L_i} \leq \sqrt{K' \cdot \sum_{i=1}^{K'} L_i} \leq \sqrt{K_{\max} T}. \quad (19)$$

The first inequality is tight when all epochs have equal length, a scenario that corresponds to an adversary who reveals types evenly across the horizon. This is where the leading multiplicative $\sqrt{K_{\max}}$ factor in the bound is generated. An instance-adaptive version uses K' in place of $K_{\max}$, which is what Corollary 8 records.

Step 10 (Final substitution). Plugging Equations (17) and (19) into Equation (18) and choosing $\epsilon = 1/\sqrt{T}$ gives

$$\begin{aligned} \mathbb{E}[\text{Regret}(T)] &\leq 2K_{\max} + 2\sqrt{T} + c_1 \sqrt{K_{\max} T \log N_{\max}} \\ &\leq 2K_{\max} + 2\sqrt{T} \\ &\quad + c_3 \sqrt{TK_{\max}(n^2 + K_{\max} \log n)} \end{aligned}$$

for an absolute constant c_3 . Whenever $T \geq K_{\max}$ (the only regime in which the statement of Theorem 6 is non-trivial) the leading $\sqrt{TK_{\max}(n^2 + K_{\max} \log n)}$ term dominates the additive $2K_{\max} + 2\sqrt{T}$, so

$$\mathbb{E}[\text{Regret}(T)] = O\left(\sqrt{T \cdot K_{\max} \cdot (n^2 + K_{\max} \log n)}\right),$$

which is exactly the bound of Theorem 6. This completes the proof. $\square$

6.1 Consequences of the proof

Two corollaries follow directly from the proof of Theorem 6 above. The first characterizes the regime in which the bound is non-trivial, the second gives an instance-adaptive refinement.

Corollary 7 (Regime of applicability). *The bound of Theorem 6 is sublinear in T if and only if $K_{\max} \cdot (n^2 + K_{\max} \log n) = o(T)$; equivalently, $K_{\max} = o(\min\{T/n^2, \sqrt{T/\log n}\})$. In particular, for $K_{\max} = O(1)$ the rate is $\tilde{O}(n\sqrt{T})$.*

Proof. Set the right-hand side of Equation (7) equal to T and solve for $K_{\max}$. The condition $K_{\max} \cdot (n^2 + K_{\max} \log n) = o(T)$ is exactly when the bound of Theorem 6 is $o(T)$; the equivalent form follows by separately bounding $K_{\max} \cdot n^2 \leq T/\omega(1)$ and $K_{\max}^2 \log n \leq T/\omega(1)$. $\square$

Corollary 8 (Instance-adaptive bound). *Let $K' = |\{a_1, \dots, a_T\}| \leq K_{\max}$ be the number of distinct attacker types actually observed in the realized sequence. Then Algorithm 1 achieves*

$$\mathbb{E}[\text{Regret}(T)] \leq O\left(\sqrt{T \cdot K' \cdot (n^2 + K' \log n)}\right).$$

Proof. The Cauchy–Schwarz step Equation (19) of the proof of Theorem 6 gives $\sum_{i=1}^{K'} \sqrt{L_i} \leq \sqrt{K' \cdot T}$; plugging this sharper bound into Equation (18) and redoing Step 10 with K' in place of $K_{\max}$ gives the stated bound. $\square$

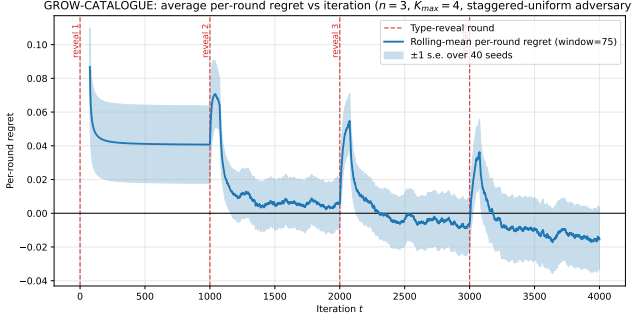


Figure 1: Rolling-window per-round regret of GROW-CATALOGUE under the staggered-uniform adversary ($n = 3$, $K_{\max} = 4$, $T = 4000$, average over 40 seeds, window = 75). Red dashed lines mark the discovery rounds 0, 1000, 2000, 3000. Each reveal causes a visible transient spike, followed by PW re-convergence inside the new epoch.

7 Experiments

7.1 Setup

We implement Algorithm 1 in Python, along with two reference algorithms: the known- K oracle of Balcan et al. [2, Sec. 5], which receives the full set K at $t = 1$; and a *uniform* baseline that plays $\mathbf{p}_{\text{def}} = (R/n, \dots, R/n)$ every round. Unless stated otherwise, we use $n = 3$ targets, $R = 1$ resource, and attacker utilities drawn i.i.d. from $\text{Unif}[-1, 1]$ with $u_{\alpha}^u(i) > 0 > u_{\alpha}^c(i)$ to ensure covering is costly for the attacker. The extreme-point set $\mathcal{E}(C; \epsilon)$ is computed by direct enumeration of intersections of $(n - 1)$ -subsets of the defining hyperplanes, which is tractable at $n = 3$.

We use a *staggered-uniform* adversary: the horizon $[0, T)$ is partitioned into $K_{\max}$ phases of length $T/K_{\max}$ each, and in phase i the attacker is drawn uniformly at random from the first i types $\{\alpha_1, \dots, \alpha_i\}$. Thus α_1 is present in every phase, α_2 from phase 2 onward, and so on. This design produces a clean sequence of discovery rounds at $t = 0, T/K_{\max}, \dots, (K_{\max} - 1)T/K_{\max}$, while still mixing old and new types after each reveal so that the benchmark $\mathbf{p}^*$ is a genuine compromise rather than a specialist strategy.

7.2 Results

Figure 1 shows that the algorithm’s per-round regret tracks the epoch structure of the proof: each discovery round produces a bounded transient, and the PW sub-routine re-converges within the new epoch. The spikes shrink with i because, by epoch i , the algorithm already has a good PW distribution on most of the new (refined) extreme points.

Figure 2 plots cumulative $\text{Regret}(T)$ versus T with the $C\sqrt{T}$ envelope from Theorem 6 overlaid. The empirical regret stays visibly below the envelope for all T tested, and the ratio $\text{Regret}(T)/\sqrt{T}$ remains bounded (in fact, it decreases, indicating that the stochastic staggered-uniform adversary is far from the worst case). This is entirely consistent with Theorem 6, which is our upper bound.

Table 1 compares the four algorithms at $T = 4000$. The regret of GROW-CATALOGUE is within roughly 15%–40% of

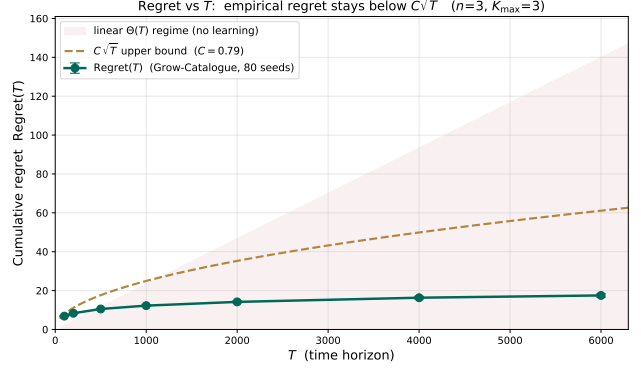


Figure 2: Cumulative regret versus horizon T for GROW-CATALOGUE ($n = 3$, $K_{\max} = 3$, uniform random adversary, 80 seeds). The green curve stays strictly below the $C\sqrt{T}$ upper envelope implied by Theorem 6, which is in turn well below the linear-regret regime that would result without learning.

the known- K oracle and an order of magnitude better than the uniform baseline. The two GROW-CATALOGUE variants produce regret that differs only by Monte-Carlo noise, as predicted by the observational equivalence of the two recomputation schemes.

Algorithm	$\mathbb{E}[\text{Regret}(T)]$	Ratio
Known- K oracle of Balcan et al. [2]	14.51 ± 5.00	1.00
GROW-CAT. (total)	16.57 ± 5.40	1.14
Uniform baseline	292.77 ± 350.28	20.18

Table 1: Cumulative regret at $T = 4000$, $n = 3$, $K_{\max} = 3$, uniform-random adversary, mean $\pm$ std over 15 seeds. The known- K oracle receives K in advance; GROW-CATALOGUE attains comparable regret *without* this information.

8 Conclusions

We studied the repeated Stackelberg security game in a new, open regime in which the set of attacker types is unknown to the defender a priori but known to have bounded cardinality. We presented GROW-CATALOGUE, a simple online algorithm that incrementally discovers attacker types and re-partitions the coverage polytope accordingly, and we gave a full proof that its expected regret under full-information feedback is at most $O(\sqrt{T} \cdot K_{\max} \cdot (n^2 + K_{\max} \log n))$. This is the same functional form as the known- K bound of Balcan et al. [2, Theorem 5.1], with the known cardinality k replaced by our known upper bound $K_{\max}$. The bound is sublinear whenever $K_{\max} \cdot (n^2 + K_{\max} \log n) = o(T)$, and the matching $\Omega(T)$ lower bound of Balcan et al. [2, Theorem 7.1] rules out further improvements on the unbounded side. Our experiments confirm the analytical scaling: the empirical regret stays below the $C\sqrt{T}$ envelope and tracks the epoch structure of the proof.

9 Partial-Information Feedback with an Unknown Type Set

Problem 4 pairs an *unknown* attacker set K with *full-information* feedback, and Theorem 6 shows that the pairing is benign: the price of not knowing K is a factor $\sqrt{K_{\max}}$ over the known- K rate of Balcan et al. [2, Theorem 5.1]. Balcan et al. [2, Section 6] show that the other relaxation is benign as well: with K *known* but only the attacked target observed, regret $O(T^{2/3}nk \log^{1/3}(nk))$ is achievable (their Theorem 6.1). It is natural to expect that the two relaxations compose, and that the unknown- K problem under partial-information feedback admits a $T^{2/3}$ -type bound with k replaced by $K_{\max}$.

This section shows that it does not, and that the failure is total: no algorithm achieves regret $o(T)$, *even when* $K_{\max} = 1$, i.e. even when a single fixed attacker type attacks in every one of the T rounds. The obstruction is not the cardinality of K — it is that the defender can never learn *which* type it is facing, because the only thing it ever observes is a target index, and the space of types that could have produced that index is infinite.

Sections 9 to 12 extend the report in four steps, and it is worth stating them together before starting. Theorem 13 is the impossibility just described. Theorem 16 sharpens it into an exact accounting: on a universe of M candidate types, and given any oracle that emits at most b bits over the whole horizon, the optimal regret is $\min\{T, \log_2 M - b\} \pm 3$, so the attainable regret is the number of bits still needed to name the attacker, and no *finite* budget helps on an infinite universe. Proposition 21 closes Section 9 by showing that this naming account is exactly a $K_{\max} = 1$ phenomenon: two attacker types already force $\Omega(\sqrt{T})$ on a two-element universe, under full information and with K known. Section 10 then fixes the model rather than the algorithm: an oracle that every B rounds reports the *set* of types met so far restores sublinear regret, at the rate $\tilde{O}(nK_{\max}^{4/3}T^{2/3} + K_{\max}B)$ (Theorem 28), with the additive term tight (Theorem 32) and with a sharp separation between scheduled and defender-chosen reporting times (Proposition 34). Section 11 makes both algorithms oracle-efficient, which moves the experiments from $n = 3$ targets to $n = 20$.

9.1 The problem

Problem 9 (Unknown- K repeated SSG, partial information). A repeated SSG as in Section 3 is played, with three modifications.

1. There is a set $\mathfrak{U}$ of attacker types, the *universe*, which is known to the defender. It is not assumed to be finite; Theorem 13 takes it infinite, while Theorem 16 and Proposition 21 measure what happens when it is not.
2. The set $K \subseteq \mathfrak{U}$ of attacker types is not revealed to the defender at the start; however, the defender is given a constant $K_{\max} \geq 1$ with the promise $|K| \leq K_{\max}$.
3. Feedback is *partial information* in the sense of Balcan et al. [2, Section 3]: at the end of round t the defender observes only the attacked target $b_{a_t}(\mathbf{p}_t)$. It never observes a_t , and the utility vectors of a type are never revealed.

The adversary is oblivious, and the benchmark $\mathbf{p}^*$ and the regret $\text{Regret}(T)$ are as in Equations (3) and (4).

Two remarks fix the exact informational content of Problem 9. First, the defender's own payoff is *not* additional feedback: the defender knows u_d^c and u_d^u , so from the observed target $i_t = b_{a_t}(\mathbf{p}_t)$ and its own $\mathbf{p}_t$ it computes $U_d(i_t, \mathbf{p}_t)$ exactly. Second, Problem 9 is at least as hard as each of the two settings it weakens: Problem 4 is recovered by revealing a_t and the utilities of each type on first contact, and the setting of Balcan et al. [2, Section 6] is recovered by revealing K at $t = 0$.

9.2 A three-target gadget

All lower bounds in this and the next section are proved in one fixed game, which we call G_3 and which is the game of Balcan et al. [2, Theorem 7.1], up to letting the single resource stand idle. It has $n = 3$ targets and one resource, with schedules $\mathcal{D} = \{\emptyset, \{1\}, \{2\}, \{3\}\}$: the resource covers one target or is left idle. The four pure strategies give $\mathcal{P} = \text{conv}\{\mathbf{0}, e_1, e_2, e_3\} = \{\mathbf{p} \in [0, 1]^3 : p_1 + p_2 + p_3 \leq 1\}$. Allowing the resource to idle only enlarges the defender's action set and therefore only strengthens the lower bounds; every argument below also goes through verbatim on the simplex $\{\mathbf{p} \geq 0 : \sum_i p_i = 1\}$ that one gets by forcing the resource to be assigned, since each of the coverage vectors we exhibit has $\sum_i p_i = 1$. The defender's payoffs are

$$u_d^c(i) = u_d^u(i) = -1 \quad (i \in \{1, 3\}), \quad u_d^c(2) = u_d^u(2) = 0, \quad (20)$$

so that by Equation (1)

$$U_d(1, \mathbf{p}) = U_d(3, \mathbf{p}) = -1, \quad U_d(2, \mathbf{p}) = 0 \quad (21)$$

for every $\mathbf{p} \in \mathcal{P}$. Throughout Sections 9 and 10 the types of G_3 break ties in favour of the lowest-indexed target, as in Balcan et al. [2, Section 7]; Section 3 permits any fixed consistent rule, and fixing one is part of specifying the instance. The choice is used only to describe the feedback: at every coverage vector we exhibit as a benchmark, the attacked target is the *strict* maximizer, so the value of the benchmark does not depend on the rule.

In G_3 the defender's payoff depends on $\mathbf{p}_t$ *only* through which target the attacker chooses: the round is worth 0 if target 2 is attacked and -1 otherwise. We say the defender *hits* at round t if $b_{a_t}(\mathbf{p}_t) = 2$ and *misses* otherwise.

We now record the family of attacker types that G_3 admits. The statement is Balcan et al. [2, Lemma 7.2]; we give the utility vectors explicitly, because the lower bounds below need the best response at *every* $\mathbf{p} \in \mathcal{P}$ and not only on the segment considered there.

Lemma 10 (realizable threshold types; cf. 2, Lemma 7.2). *For every pair $0 \leq r_1 \leq r_2 \leq 1$, the attacker type α_{r_1, r_2} with utility vectors*

$$u_\alpha^c = (r_1 - 1, 0, -r_2), \quad u_\alpha^u = (r_1, 0, 1 - r_2) \quad (22)$$

has all its utilities in $[-1, 1]$ and satisfies, for every $\mathbf{p} \in \mathcal{P}$,

$$U_\alpha(1, \mathbf{p}) = r_1 - p_1, \quad U_\alpha(2, \mathbf{p}) = 0, \quad U_\alpha(3, \mathbf{p}) = (1 - r_2) - p_3. \quad (23)$$

Consequently, if ties are broken in favour of the lowest-indexed target, the restriction of b_α to the segment $\{(p, 0, 1 - p) : p \in [0, 1]\}$ is

$$b_{\alpha_{r_1, r_2}}((p, 0, 1 - p)) = \begin{cases} 1 & p \in [0, r_1], \\ 2 & p \in (r_1, r_2], \\ 3 & p \in (r_2, 1]. \end{cases} \quad (24)$$

Proof. Since $0 \leq r_1 \leq r_2 \leq 1$ we have $r_1 - 1 \in [-1, 0]$, $-r_2 \in [-1, 0]$, $r_1 \in [0, 1]$ and $1 - r_2 \in [0, 1]$, so all six numbers in Equation (22) lie in $[-1, 1]$ and α_{r_1, r_2} is a legitimate attacker type in the sense of Section 3. For Equation (23), substitute Equation (22) into $U_\alpha(i, \mathbf{p}) = u_\alpha^c(i)p_i + u_\alpha^u(i)(1 - p_i)$:

$$\begin{aligned} U_\alpha(1, \mathbf{p}) &= (r_1 - 1)p_1 + r_1(1 - p_1) = r_1 - p_1, \\ U_\alpha(2, \mathbf{p}) &= 0 \cdot p_2 + 0 \cdot (1 - p_2) = 0, \\ U_\alpha(3, \mathbf{p}) &= -r_2p_3 + (1 - r_2)(1 - p_3) = (1 - r_2) - p_3. \end{aligned}$$

On the segment $\mathbf{p} = (p, 0, 1 - p)$ these read $U_\alpha(1, \mathbf{p}) = r_1 - p$, $U_\alpha(2, \mathbf{p}) = 0$ and $U_\alpha(3, \mathbf{p}) = p - r_2$. If $p \leq r_1$ then $U_\alpha(1, \mathbf{p}) \geq 0$ and $U_\alpha(3, \mathbf{p}) = p - r_2 \leq r_1 - r_2 \leq 0$, so target 1 attains the maximum (and wins any tie, being lowest-indexed). If $r_1 < p \leq r_2$ then $U_\alpha(1, \mathbf{p}) < 0$ and $U_\alpha(3, \mathbf{p}) \leq 0 = U_\alpha(2, \mathbf{p})$, so target 2 attains the maximum and wins the tie against target 3. If $p > r_2$ then $U_\alpha(3, \mathbf{p}) > 0 = U_\alpha(2, \mathbf{p}) > U_\alpha(1, \mathbf{p})$, and target 3 is the strict maximizer. $\square$

The next lemma is the structural fact that drives both lower bounds. It says that in G_3 a hit is a very informative but very rare event: a coverage vector hits at most one member of any family of types with disjoint threshold intervals.

Lemma 11 (the hit condition). *Let $0 \leq r_1 < r_2 \leq 1$ and write $\alpha = \alpha_{r_1, r_2}$. For every $\mathbf{p} \in \mathcal{P}$,*

$$b_\alpha(\mathbf{p}) = 2 \iff p_1 > r_1 \text{ and } p_3 \geq 1 - r_2, \quad (25)$$

and in that case $p_1 \in (r_1, r_2]$. Consequently, if $\{\alpha_{r_1^{(j)}, r_2^{(j)}}\}_{j \in J}$ is any family of types whose intervals $(r_1^{(j)}, r_2^{(j)})$ are pairwise disjoint, then for every $\mathbf{p} \in \mathcal{P}$ there is at most one $j \in J$ with $b_{\alpha_{r_1^{(j)}, r_2^{(j)}}}(\mathbf{p}) = 2$.

Proof. By Equation (23) and lowest-index tie-breaking, target 2 is chosen exactly when $U_\alpha(2, \mathbf{p}) = 0$ strictly beats $U_\alpha(1, \mathbf{p})$ and weakly beats $U_\alpha(3, \mathbf{p})$, that is, when $r_1 - p_1 < 0$ and $(1 - r_2) - p_3 \leq 0$; this is Equation (25). If it holds, then $p_1 \leq 1 - p_2 - p_3 \leq 1 - p_3 \leq r_2$ because $\mathbf{p} \in \mathcal{P}$ and $p_2 \geq 0$, so $p_1 \in (r_1, r_2]$. The last claim follows because a single p_1 cannot lie in two disjoint intervals. $\square$

9.3 Identification is a comparison search

Fix an integer $M \geq 2$ and let

$$I_j = \left(\frac{j-1}{M}, \frac{j}{M} \right], \quad \alpha^{(j)} = \alpha_{\frac{j-1}{M}, \frac{j}{M}}, \quad j \in [M], \quad (26)$$

so that $I_1, \dots, I_M$ partition $(0, 1]$ and $\mathcal{U}_M := \{\alpha^{(1)}, \dots, \alpha^{(M)}\}$ is a set of M distinct attacker types. Lemma 11 applies to $\mathcal{U}_M$ because the I_j are pairwise disjoint.

The defender's whole interaction with an instance $K = \{\alpha^{(j)}\}$ is then a search for the index j in which every query

returns one of three answers and the answer “2” is consistent with at most one candidate. The following lemma bounds how fast *any* such search can succeed. It refines the classical $\Omega(\log_2 M)$ bound for comparison-based search in three ways that the statements below need: it charges *every* unsuccessful query rather than only counting queries until termination; it bounds the whole tail of the success time rather than its worst case; and it allows the searcher to be fed, alongside the answers, an arbitrary side channel of bounded total width. It is the combinatorial core of Theorems 13, 16 and 32, and is stated abstractly so that it can be used three times.

Lemma 12 (comparison trees identify slowly). *Let $M \geq 1$, let j be drawn uniformly at random from $[M]$, and fix a side-information budget $b \geq 0$. Consider any interactive protocol in which, for $t = 1, 2, \dots$, a deterministic strategy picks a query q_t as a function of everything it has seen so far and then receives, in this order, an answer $o_t = \mathcal{O}(q_t, j) \in \{1, 2, 3\}$ and a side message $\mu_t \in \Sigma_t$. The alphabets $\Sigma_0, \Sigma_1, \Sigma_2, \dots$ are fixed before play, $\mu_0 \in \Sigma_0$ is delivered before the first query, each μ_t may be an arbitrary function of j and of the history, and the total width of the side channel is bounded:*

$$\sum_{t \geq 0} \log_2 |\Sigma_t| \leq b. \quad (27)$$

Assume the single requirement that

$$|\{j' \in [M] : \mathcal{O}(q, j') = 2\}| \leq 1 \quad \text{for every query } q. \quad (28)$$

Let $\tau = \min\{t : o_t = 2\}$, with $\tau = \infty$ if no such t exists. Then

$$\Pr[\tau \leq u] \leq \frac{2^b(2^u - 1)}{M} \quad \text{for every } u \geq 1, \quad (29)$$

and consequently, for every $T \geq 1$,

$$\mathbb{E}[\min\{\tau - 1, T\}] \geq T - 2^b \frac{2^{T+1} - T - 2}{M} \quad (30)$$

and, in the cruder form that is easiest to read off,

$$\mathbb{E}[\min\{\tau - 1, T\}] \geq \min\{T, \lfloor \log_2 M \rfloor - b\} - 2. \quad (31)$$

Taking $b = 0$, i.e. $|\Sigma_t| = 1$ for every t , gives the side-channel-free statement used in Theorems 13 and 32.

Proof. Fix $t \geq 1$ and let H_t be the set of transcripts $(\mu_0, o_1, \mu_1, \dots, o_{t-1}, \mu_{t-1})$ of the first $t-1$ rounds that contain no answer 2. Each such transcript has $o_s \in \{1, 3\}$ for $s \leq t-1$ and $\mu_s \in \Sigma_s$ for $s \leq t-1$, so by Equation (27)

$$|H_t| \leq 2^{t-1} \prod_{s=0}^{t-1} |\Sigma_s| \leq 2^{t-1} 2^b.$$

Suppose $\tau(j) = t$. Then the first $t-1$ rounds produced by j form some transcript $h \in H_t$, and, since the strategy is deterministic, the query issued at round t is the fixed query $q_t(h)$ determined by h ; moreover $\mathcal{O}(q_t(h), j) = 2$. By Equation (28) at most one index answers 2 to $q_t(h)$, so for each $h \in H_t$ there is at most one j with $\tau(j) = t$ and transcript h . As distinct j with $\tau(j) = t$ must be counted under some $h \in H_t$,

$$|\{j \in [M] : \tau(j) = t\}| \leq |H_t| \leq 2^b 2^{t-1},$$

and summing over $t \leq u$ gives $|\{j : \tau(j) \leq u\}| \leq 2^b \sum_{t=1}^u 2^{t-1} = 2^b(2^u - 1)$. Dividing by M proves Equation (29).

For Equation (30), put $X = \min\{\tau - 1, T\}$, a random variable with values in $\{0, 1, \dots, T\}$. For $0 \leq s \leq T - 1$ we have $X > s$ if and only if $\tau - 1 > s$, i.e. $\tau > s + 1$. Hence

$$\begin{aligned} \mathbb{E}[X] &= \sum_{s=0}^{T-1} \Pr[X > s] = \sum_{u=1}^T \Pr[\tau > u] = \sum_{u=1}^T (1 - \Pr[\tau \leq u]) \\ &\geq T - 2^b \sum_{u=1}^T \frac{2^u - 1}{M} = T - 2^b \frac{2^{T+1} - 2 - T}{M}, \end{aligned}$$

using $\sum_{u=1}^T 2^u = 2^{T+1} - 2$.

For Equation (31), write $L = \lfloor \log_2 M \rfloor - b$. If $L \leq 0$ the claim is vacuous because $X \geq 0$. If $0 < L$ and $T \leq L$, then $M \geq 2^{\lfloor \log_2 M \rfloor} = 2^{L+b}$, so the subtracted term in Equation (30) is at most $2^b 2^{T+1} / 2^{L+b} = 2^{T+1-L} \leq 2$, giving $\mathbb{E}[X] \geq T - 2$. If $0 < L < T$, let $h = \lfloor L \rfloor$ and $f = L - h \in [0, 1)$. Then $h \leq T$ is an integer and $X \geq \min\{\tau - 1, h\}$ pointwise, so Equation (30) with horizon h , together with $M \geq 2^{L+b}$, gives $\mathbb{E}[X] \geq h - 2^b 2^{h+1} / M = h - 2^{1-f}$. Now $2 - f \geq 2^{1-f}$ for every $f \in [0, 1)$: the two sides agree at $f = 0$, and on $[0, 1)$ the left side falls with slope -1 while the right side falls with slope $-2^{1-f} \ln 2 \leq -\ln 2 > -1$. Hence $\mathbb{E}[X] \geq h + f - 2 = L - 2$. In all cases $\mathbb{E}[X] \geq \min\{T, L\} - 2$; when b is an integer and M a power of two we have $f = 0$ and the middle step is the plain $2^{T+1-L} \leq 2$ of the previous case. $\square$

9.4 The impossibility theorem

Theorem 13 (unknown K with partial information is not learnable). *Let G_3 be the game of Equation (20) and let $\mathfrak{U} = \{\alpha_{r_1, r_2} : 0 \leq r_1 \leq r_2 \leq 1\}$ be the (infinite) universe of threshold types of Lemma 10. Then for every horizon $T \geq 1$ and every $K_{\max} \geq 1$, every online algorithm for Problem 9 — randomized or deterministic, and even one that knows $T, \mathfrak{U}, K_{\max}$ and the construction below — suffers, on some instance with $|K| = 1$,*

$$\mathbb{E}[\text{Regret}(T)] \geq T - 2 + \frac{T+2}{2^T} > T - 2. \quad (32)$$

The game G_3 does not depend on T ; only the adversary's choice of K inside the fixed universe $\mathfrak{U}$ does.

Proof. Set $M = 2^T$ and let $\mathfrak{U}_M = \{\alpha^{(1)}, \dots, \alpha^{(M)}\} \subset \mathfrak{U}$ be as in Equation (26). Following the route of Balcan et al. [2, Theorem 7.1], we invoke Yao's minimax principle: we exhibit a distribution over instances and lower bound the expected regret of every *deterministic* algorithm against it.

The distribution over instances. Draw j uniformly from $[M]$ and set $K = \{\alpha^{(j)}\}$ and $a_t = \alpha^{(j)}$ for every $t \in [T]$. This is a legal instance of Problem 9 for every $K_{\max} \geq 1$, since $|K| = 1$; the sequence $\mathbf{a}$ is fixed before play begins, so the adversary is oblivious.

Step 1 (the benchmark is 0). Let $\mathbf{p}^* = (\frac{2j-1}{2M}, 0, 1 - \frac{2j-1}{2M})$, whose first coordinate is the midpoint of I_j and which lies in $\mathcal{P}$. With $r_1 = \frac{j-1}{M}$ and $r_2 = \frac{j}{M}$ we have $p_1^* = \frac{2j-1}{2M} > \frac{2j-2}{2M} = r_1$ and $p_3^* = 1 - \frac{2j-1}{2M} > 1 - \frac{2j}{2M} = 1 - r_2$, both strictly, so Equation (25) gives $b_{\alpha^{(j)}}(\mathbf{p}^*) = 2$ — and does so under *any* tie-breaking rule, since target 2 is the strict maximizer at $\mathbf{p}^*$. Hence, by Equation (21), $U_d(b_{a_t}(\mathbf{p}^*), \mathbf{p}^*) = 0$ for every t . Since $U_d(i, \mathbf{p}) \leq 0$ for every target i and every $\mathbf{p}$ by Equation (21), no coverage vector can do better, so this $\mathbf{p}^*$ is indeed a maximizer in Equation (3). Therefore $\sum_{t=1}^T U_d(b_{a_t}(\mathbf{p}^*), \mathbf{p}^*) = 0$, and by Equation (4)

$$\begin{aligned} \text{Regret}(T) &= -\mathbb{E}\left[\sum_{t=1}^T U_d(b_{a_t}(\mathbf{p}_t), \mathbf{p}_t)\right] \\ &= \mathbb{E}[\#\{t \leq T : \text{miss at } t\}], \end{aligned} \quad (33)$$

again by Equation (21). So regret counts missed rounds exactly.

Step 2 (the interaction is a comparison search). A deterministic algorithm for Problem 9 is a family of maps $\mathbf{p}_t = \mathbf{p}_t(o_1, \dots, o_{t-1})$, where $o_s = b_{a_s}(\mathbf{p}_s) \in \{1, 2, 3\}$ is the only feedback it receives; its own past plays are determined by the same history and carry no extra information, and as noted after Problem 9 its realized payoffs are functions of $(o_s, \mathbf{p}_s)$. Define the query $q_t = \mathbf{p}_t$ and the answer map $\mathcal{O}(\mathbf{p}, j') = b_{\alpha^{(j')}}(\mathbf{p})$. The intervals $I_1, \dots, I_M$ are pairwise disjoint, so Lemma 11 gives exactly the hypothesis Equation (28): for every $\mathbf{p} \in \mathcal{P}$ at most one $j' \in [M]$ satisfies $\mathcal{O}(\mathbf{p}, j') = 2$. Hence the interaction is a protocol of the type covered by Lemma 12, with budget $b = 0$ because Problem 9 gives the defender no channel other than o_t .

Step 3 (counting). Let τ be the first round at which the algorithm hits. Every round $t < \tau$ is a miss, and there are at least $\min\{\tau - 1, T\}$ such rounds in $[T]$. Combining with Equation (33) and Equation (30) with $b = 0$ and $M = 2^T$,

$$\begin{aligned} \mathbb{E}_j[\text{Regret}(T)] &\geq \mathbb{E}_j[\min\{\tau - 1, T\}] \\ &\geq T - \frac{2^{T+1} - T - 2}{2^T} = T - 2 + \frac{T+2}{2^T}. \end{aligned}$$

Step 4 (Yao). Let A be any randomized algorithm and let A_ω denote its deterministic realization under internal randomness ω . Then

$$\begin{aligned} \max_{K, \mathbf{a}} \mathbb{E}[\text{Regret}(T; A)] &\geq \mathbb{E}_j \mathbb{E}_\omega[\text{Regret}(T; A_\omega, j)] \\ &= \mathbb{E}_\omega \mathbb{E}_j[\text{Regret}(T; A_\omega, j)], \end{aligned}$$

and the inner expectation is at least $T - 2 + (T+2)2^{-T}$ for every ω by Steps 1–3. This proves Equation (32). $\square$

Remark 14 (the bound is exactly tight). The constant in Equation (32) cannot be improved. The defender that maintains the set of indices consistent with the observations so far — an integer interval, by Equation (24) — and plays $\mathbf{p}_t = (k/M, 0, 1 - k/M)$ for its midpoint k attains expected regret exactly $T - 2 + (T+2)2^{-T}$ on this family, as we verify in Section 11.5. Since $\text{Regret}(T) \leq T$ always in G_3 , Theorem 13 says the defender loses all but two of the T rounds.

9.5 What exactly fails, and what would fix it

Theorem 13 is not a statement about how many types the adversary has; the hard instance uses a single type, deployed in every round. It is a statement about how many types the adversary *could* have had. The next theorem makes that precise by re-running the argument on a universe of finite resolution, exhibiting a matching algorithm, and — since the diagnosis is that the defender is missing the *name* of a type — pricing the supplement that would repair it. We charge that supplement in bits.

Definition 15 (side-information oracle of budget b). A *side-information oracle of budget b* for Problem 9 is given by alphabets $\Sigma_0, \Sigma_1, \dots, \Sigma_T$ fixed before play with $\sum_t \log_2 |\Sigma_t| \leq b$, together with message maps that hand the defender $\mu_0 \in \Sigma_0$ before round 1 and $\mu_t \in \Sigma_t$ at the end of round t . Each μ_t may be an arbitrary function of the instance $(K, \mathbf{a})$ and of the history so far. Nothing is assumed about what the messages mean; b measures only how many bits of them there are.

Theorem 16 (regret is the number of bits needed to name the type). *Fix $M \geq 2$ and a budget $b \geq 0$, and consider Problem 9 in G_3 with $K_{\max} = 1$, augmented by a side-information oracle of budget b . Parts (i) and (iii) hold for every real $b \geq 0$; part (ii), which has to build 2^b blocks, is stated for integer b .*

1. *On the universe $\mathfrak{U}_M$ of Equation (26), every algorithm and every such oracle suffer $\mathbb{E}[\text{Regret}(T)] \geq \min\{T, \lfloor \log_2 M \rfloor - b\} - 2$ on some instance.*
2. *For integer b , on $\mathfrak{U}_M$ there are an oracle of budget b and an algorithm that together suffer $\text{Regret}(T) \leq \min\{T, \max\{0, \lfloor \log_2 M \rfloor - b\}\}$ on every instance.*
3. *On the infinite universe $\mathfrak{U}$ of Theorem 13, every algorithm and every oracle of any finite budget b suffer $\mathbb{E}[\text{Regret}(T)] \geq T - 2$ on some instance with $|K| = 1$.*

Hence on $\mathfrak{U}_M$ the optimal regret is $\min\{T, \log_2 M - b\} \pm 3$ — and ± 2 when M is a power of two, which is the only case used below — so one bit of threat intelligence buys exactly one round, until the type has been named and the remaining rounds are free. Taking $b = 0$ recovers the statement that the optimal regret is $\min\{T, \log_2 |\mathfrak{U}_M|\} \pm 3$, and (iii) says that on an infinite universe no fixed quantity of intelligence buys anything at all.

Proof. (i) Repeat Steps 1–4 of the proof of Theorem 13 on $\mathfrak{U}_M$ verbatim; the only change is in Step 2, where the protocol handed to Lemma 12 now also carries the messages μ_t , so it is a protocol of budget b rather than 0. Note that Lemma 12 allows q_t to depend on $\mu_0, \dots, \mu_{t-1}$, which is exactly the freedom a defender reading the oracle has, and that Equation (28) is a property of the answer map $\mathcal{O}$ alone and is therefore untouched by the side channel. Step 3 now invokes Equation (31) instead of Equation (30) and gives $\mathbb{E}_j[\text{Regret}(T)] \geq \min\{T, \lfloor \log_2 M \rfloor - b\} - 2$ for every deterministic algorithm; Step 4 is unchanged.

(ii) Write $m = \lfloor \log_2 M \rfloor$. If $b \geq m$ let the oracle name j outright, using $\lfloor \log_2 M \rfloor \leq b$ bits, and let the defender play the midpoint of I_j forever; by Equation (25) it hits in every round and the regret is 0. Otherwise split $[M]$ into 2^b consecutive blocks of $\lceil M/2^b \rceil$ or fewer indices each and let μ_0 name the block containing j ; this is one message from an alphabet of size 2^b , so the budget is met. The defender then runs the midpoint

search of Remark 14 restricted to that block: it keeps the sub-interval $[\ell, h]$ of indices consistent with everything it has seen, initially the announced block, and plays $\mathbf{p} = (k/M, 0, 1 - k/M)$ for its midpoint k . By Equation (24) at that $\mathbf{p}$, the observation is target 2 if $j = k$, target 1 if $j > k$ and target 3 if $j < k$, so every missed round replaces $[\ell, h]$ by one of its two halves and every hit leaves it unchanged. Starting from $\lceil M/2^b \rceil \leq 2^{m-b}$ candidates, at most $m - b$ misses can occur before $[\ell, h]$ is a singleton, after which every round is a hit; and trivially at most T rounds are missed. By Equation (33) the regret equals the number of misses.

(iii) Apply (i) on the sub-universe $\mathfrak{U}_M \subset \mathfrak{U}$ with $M = 2^{T+b}$, for which $\lfloor \log_2 M \rfloor - b = T$; the instance produced has $|K| = 1$. $\square$

Remark 17 (what the budget does and does not cover). Definition 15 is deliberately permissive: the oracle may look at the whole instance, may speak before the game starts, may adapt its messages to the defender’s plays, and is free to use any encoding at all. What it may not do is exceed b bits in total. The lower bound is therefore information-theoretic and not an artefact of any particular reporting format; in particular it applies to the block-revelation oracle of Section 10 whenever that oracle’s messages can be encoded in b bits. What Theorem 16 does *not* capture is *when* the bits arrive: Theorem 32 below shows that even an oracle with a generous budget still costs $\Theta(K_{\max} B)$ if it is made to speak only every B rounds. Volume and latency are two independent obstructions, and the two lower bounds isolate them one at a time.

Remark 18 (the two lower bounds are incomparable). Table 2 places Theorem 13 among the known results. Balcan et al. [2, Theorem 7.1] obtain $\Omega(T)$ with *full* information, but their construction uses a type set of size $2^{T+1} - 2$, and the sequence it plays — the T postfixes of a random T -bit string — contains T *distinct* types, so it needs $K_{\max} \geq T$ and is consistent with Theorem 6. Theorem 13 obtains $\Omega(T)$ with $|K| = 1$, and hence for every $K_{\max} \geq 1$, but needs partial information and an infinite universe. Neither implies the other. Together with Theorem 6 and Balcan et al. [2, Theorem 6.1] they show that either relaxation of Balcan et al. [2, Theorem 5.1] alone is affordable — a $\sqrt{K_{\max}}$ multiplicative price for not knowing K , an exponent $2/3$ instead of $1/2$ for not observing a_t — while their conjunction is not affordable at all.

Remark 19 (the adversary is not the difficulty). The instance of Theorem 13 is stationary: one fixed type attacks in every round. The lower bound therefore continues to hold if the defender is additionally told $|K|$ exactly, told the full frequency vector of the type sequence, or told that the sequence is i.i.d.; none of these is informative when $|K| = 1$. What the defender is missing is not information about *which types attack when*, which is the quantity that Balcan et al. [2, Section 6] must estimate, but information about the identity of the types themselves.

Remark 20 (reconciliation with query-based learning). Letchford et al. [14] show that a defender facing a *single* unknown attacker type can learn a near-optimal commitment from a small number of best-response queries, which at first sight is in tension with Theorem 13. There is no contradiction. Any query bound of that kind must depend on the geometry

Feedback	Set K	Regret
full	known, $ K = k$	$O(\sqrt{Tn^2k \log nk})$
partial	known, $ K = k$	$O(T^{2/3}nk \log^{1/3}nk)$
full	known, $ K = 2^{T+1} - 2$	$\Omega(T)$
full	unknown, $\leq K_{\max}$	$O(\sqrt{TK_{\max}\Lambda_n})$
partial	unknown, $\leq K_{\max}$	$\Omega(T)$, even $K_{\max}=1$
partial	unknown, $K_{\max}=1$, $ \mathcal{U} =M$, b bits	$\min\{T, \log_2 M - b\} \pm 3$
full	known, $ K = \mathcal{U} =2$	$\Theta(\sqrt{T})$
partial	unknown, $\leq K_{\max}$, block revelation	$\tilde{O}(nK_{\max}^{4/3}T^{2/3} + K_{\max}B)$

Table 2: The landscape, with $\Lambda_n = n^2 + K_{\max} \log n$. Rows 1–3 are Theorems 5.1, 6.1 and 7.1 of Balcan et al. [2]; row 4 is Theorem 6; row 5 is Theorem 13; row 6 is Theorem 16, which is tight up to an additive 3; row 7 is Proposition 21 together with Theorem 6 at $k = 2$, and is tight up to a constant. The $\Omega(\sqrt{T})$ half of row 7 holds under partial information as well, but the matching upper bound does not: with partial information the best bound we can cite for a known two-type set is the $O(T^{2/3})$ of Balcan et al. [2, Theorem 6.1], and closing that gap is one of the open problems of Section 12. Row 8 is Theorem 28, in the model of Problem 23.

of the best-response regions, and the family $\mathcal{U}_M$ of Equation (26) drives the region that has to be located down to width $1/M = 2^{-T}$. Theorem 16 says exactly that a dependence of this kind cannot be removed: the attainable regret is the number of bits needed to name the attacker type, and an infinite universe makes that number unbounded.

9.6 A $\sqrt{T}$ floor as soon as $K_{\max} \geq 2$

Theorem 16 makes the $K_{\max} = 1$ problem entirely a naming problem: on a universe of resolution M the regret is $\min\{T, \log_2 M - b\} \pm 3$, which for a fixed finite universe does not grow with the horizon at all. The next proposition shows that this is a property of $K_{\max} = 1$ and of nothing else. With two attacker types the regret is $\Omega(\sqrt{T})$ on the *smallest possible* universe, under *full-information* feedback, and even if the defender is told K in advance. Two consequences are worth naming. First, the $\sqrt{T}$ of Theorem 6 is not an artefact of the analysis. Second, whatever the answer to the finite- M question of Section 12 turns out to be for $K_{\max} \geq 2$, it must keep growing with the horizon, and so it cannot have the saturating form $\min\{T, f(M, K_{\max})\}$ that governs $K_{\max} = 1$.

Proposition 21 (two types already cost $\sqrt{T}$). *Consider G_3 with the two attacker types $\alpha^{(1)} = \alpha_{0,1/2}$ and $\alpha^{(2)} = \alpha_{1/2,1}$ of Lemma 10, so that $|\mathcal{U}| = |K| = K_{\max} = 2$. Let $a_1, \dots, a_T$ be drawn i.i.d. and uniformly from K . Then every online algorithm — randomized or deterministic, under full information or partial information, and even one that is told K together with the utility vectors at $t = 0$ — satisfies*

$$\mathbb{E}[\text{Regret}(T)] \geq \frac{1}{2}\sqrt{T/2} \quad (34)$$

for every $T \geq 2$, where the expectation is over the sequence as well as the algorithm’s randomization; consequently some fixed sequence forces at least this much.

Proof. The intervals $(0, \frac{1}{2}]$ and $(\frac{1}{2}, 1]$ of the two types are disjoint, so Lemma 11 applies: every $\mathbf{p} \in \mathcal{P}$ hits at most one of $\alpha^{(1)}, \alpha^{(2)}$. Write $N_i = |\{t : a_t = \alpha^{(i)}\}|$, so $N_1 + N_2 = T$.

The benchmark. A fixed $\mathbf{p}$ earns 0 on the rounds at which it hits and -1 on the others, by Equation (21), and by the previous paragraph it hits on at most $\max\{N_1, N_2\}$ rounds. That value is attained: $\mathbf{p} = (\frac{1}{4}, 0, \frac{3}{4})$ satisfies $p_1 > 0$ and $p_3 \geq \frac{1}{2}$, so by Equation (25) it hits $\alpha^{(1)}$ in every one of its N_1 rounds, and symmetrically $\mathbf{p} = (\frac{3}{4}, 0, \frac{1}{4})$ hits $\alpha^{(2)}$ in every one of its N_2 rounds. Hence $\sum_t U_d(b_{a_t}(\mathbf{p}^*), \mathbf{p}^*) = -\min\{N_1, N_2\}$.

The algorithm. Fix a round t and condition on the algorithm’s history, which is a function of $a_1, \dots, a_{t-1}$ and of its internal randomization and is therefore independent of a_t . The played $\mathbf{p}_t$ hits at most one of the two types, and a_t equals that type with probability $\frac{1}{2}$; so $\Pr[\text{hit at } t] \leq \frac{1}{2}$ conditionally, hence unconditionally. Summing, $\mathbb{E}[\sum_t U_d(b_{a_t}(\mathbf{p}_t), \mathbf{p}_t)] = -\mathbb{E}[\#\text{misses}] \leq -T/2$.

Combining. Since $\min\{N_1, N_2\} = \frac{1}{2}(T - |N_1 - N_2|)$,

$$\mathbb{E}[\text{Regret}(T)] \geq \frac{T}{2} - \mathbb{E}[\min\{N_1, N_2\}] = \frac{1}{2}\mathbb{E}[|N_1 - N_2|].$$

Now $N_1 - N_2 = S_T := \sum_{t \leq T} \varepsilon_t$ with $\varepsilon_t = \pm 1$ independent and uniform. The process $|S_m|$ is a submartingale, so $\mathbb{E}|S_T| \geq \mathbb{E}|S_{2\nu}|$ with $\nu = \lfloor T/2 \rfloor$, and the classical identity $\mathbb{E}|S_{2\nu}| = 2\nu \binom{2\nu}{\nu} 4^{-\nu}$ together with $\binom{2\nu}{\nu} 4^{-\nu} \geq 1/(2\sqrt{\nu})$ gives $\mathbb{E}|S_T| \geq \sqrt{\nu}$. (The inequality $\binom{2\nu}{\nu} 4^{-\nu} \geq 1/(2\sqrt{\nu})$ holds with equality at $\nu = 1$ and follows by induction, since the ratio of consecutive left-hand sides is $(2\nu+1)/(2\nu+2)$ and that of right-hand sides is $\sqrt{\nu/(\nu+1)}$, and $(2\nu+1)^2(\nu+1) \geq \nu(2\nu+2)^2$ reduces to $\nu+1 \geq 0$.) This is Equation (34). Nothing in the argument refers to the feedback model or to knowledge of K : the only facts used are that $\mathbf{p}_t$ cannot hit both types and that a_t is independent of the past. $\square$

The bound is tight on this instance, and in a strong sense: the inequality $\Pr[\text{hit at } t] \leq \frac{1}{2}$ is an equality for any algorithm that always plays inside one of the two intervals, so every such algorithm — including follow the leader, and including one that is simply told K — has expected regret *exactly* $\frac{1}{2}\mathbb{E}|N_1 - N_2|$, which is $\sqrt{T/(2\pi)}(1 + o(1))$. The remaining freedom buys nothing at all. Figure 5(c) shows the measured curve sitting on that value.

Remark 22 (the phase transition at $K_{\max} = 2$). Theorem 16 and Proposition 21 bracket the finite-universe problem from two sides. With $K_{\max} = 1$ the regret is $\min\{T, \log_2 M\} \pm 3$ and stops growing once the type has been named. With $K_{\max} = 2$ it is $\Omega(\sqrt{T})$ even at $M = 2$, because the defender must now track which of its two known opponents is currently more frequent, and no naming can settle that. The transition is therefore not about how hard it is to identify a type but about whether identification is the whole problem. On the upper side, Balcan et al. [2, Theorem 6.1] run with $k = M$ — legitimate here, since a finite universe is a known type set — gives $O(\min\{T, T^{2/3}nM \log^{1/3}(nM)\})$ under partial information, and Theorem 6 gives $O(\sqrt{TK_{\max}(n^2 + K_{\max} \log n)})$ under full information. Closing the gap between $\Omega(\sqrt{T})$ and these is the finite- M question of Section 12.

10 Block Revelation: Recovering Sublinear Regret

Theorem 13 says that the defender must be told *something* about the identity of its opponents. Theorem 16 says how little may suffice: what the defender is missing is the name of a type, and a name is a bounded object. This section studies the weakest such supplement we could find that is realistic in the applications of Section 2 — a periodic threat-intelligence report — and shows that it restores sublinear regret at the partial-information rate of Balcan et al. [2, Theorem 6.1].

The supplement is deliberately meagre. Every B rounds, an oracle hands the defender the *set* of attacker types that have attacked at least once so far, together with their utility vectors. It does not say when any of them attacked, how often, or how many are still to come. In particular the defender still cannot attribute any individual round to a type, which is precisely the difficulty that Balcan et al. [2, Section 6] solve by estimating type frequencies from best responses; that difficulty is untouched, and it is why the rate below is $T^{2/3}$ rather than $\sqrt{T}$.

10.1 The model

Problem 23 (Unknown- K repeated SSG with block revelation). Fix a *revelation length* $B \in \mathbb{N}$, known to the defender, and partition the horizon into $N_B = \lceil T/B \rceil$ *revelation blocks*

$$\mathcal{B}_\nu = \{(\nu - 1)B + 1, \dots, \min\{\nu B, T\}\}, \quad \nu \in [N_B]. \quad (35)$$

The game is Problem 9, with one addition: at the end of block $\mathcal{B}_\nu$ the defender is given the set

$$D_\nu = \{a_t : t \leq \min\{\nu B, T\}\} \subseteq K \quad (36)$$

of types that have attacked so far, together with the utility vectors (u_α^c, u_α^u) of each $\alpha \in D_\nu$. We set $D_0 = \emptyset$. Nothing else is revealed: not $|K|$, not the rounds at which any type attacked, and not the number of times any type attacked. Within a round the feedback is still partial information.

Three observations delimit Problem 23. First, it is strictly weaker than Problem 4: even at $B = 1$ the defender learns the type of round t only when that type is new, and on every other round it still sees nothing but a target. Second, $B \geq T$ makes the revelation useless and recovers Problem 9, so by Theorem 13 some bound on B is necessary; Theorem 32 below quantifies this. Third, the oracle is *cumulative* rather than incremental — it reports the whole set D_ν , not the increment $D_\nu \setminus D_{\nu-1}$ — but the two are equivalent for a defender that remembers $D_{\nu-1}$, and we use whichever is more convenient.

10.2 The algorithm

Algorithm 2 is Algorithm 1 with two changes forced by the model: the catalogue is refreshed at checkpoints rather than at discovery rounds, and the inner learner is the partial-information algorithm of Balcan et al. [2, Algorithm 1] rather than Polynomial Weights, because the loss of an expert can no longer be computed from the feedback. We write $\mathcal{A}(C)$ for a fresh instance of that algorithm run on the ϵ -approximate extreme-point set $\mathcal{E}(C; \epsilon)$ with attacker set C ; as there, $\mathcal{A}(C)$

spends $|C|$ rounds of each of its internal estimation blocks playing the barycentric-spanner strategies and the rest playing the strategy proposed by its full-information sub-routine. We fix $\epsilon = 1/T$ once and for all rather than tuning it to a segment length. This is admissible for every segment, because the discretization error of Lemma 1 is $2\epsilon L$ and so only decreases with ϵ , while the size bound of Lemma 2 does not depend on ϵ at all; and it is convenient, because it lets a single expert set serve a catalogue for the whole horizon.

One property of $\mathcal{A}(C)$ that the proof of Theorem 6 obtained for free has to be arranged explicitly here. There the inner learner was Polynomial Weights, which by Proposition 3 is *anytime*, so the fact that an epoch's length is revealed only when the epoch ends cost nothing. The algorithm of Balcan et al. [2, Theorem 6.1], by contrast, tunes its number of estimation blocks to a horizon supplied in advance, and the epochs of Definition 25 again have lengths that the defender does not know when they begin. The following lemma supplies the anytime form that Theorem 28 will use. Its proof takes Balcan et al. [2, Theorem 6.1] as a black box except for one structural feature of the proof of their Lemma 6.2, which we quote rather than repeat.

Lemma 24 (an anytime partial-information learner). *Let C be a finite catalogue, let T be the horizon of Problem 23, and let $\epsilon \leq 1/T$. There is an algorithm $\mathcal{A}(C)$ for the partial-information setting with known attacker set C , running over $\mathcal{E}(C; \epsilon)$ and not receiving its own horizon in advance, and an absolute constant c_2 (distinct from the c_1 of Section 6), such that for every oblivious sequence $a'_1, a'_2, \dots$ with $a'_t \in C$ for all t and for every $1 \leq L \leq T$ simultaneously,*

$$\begin{aligned} \max_{\mathbf{p} \in \mathcal{P}} \sum_{t=1}^L U_d(b_{a'_t}(\mathbf{p}), \mathbf{p}) - \mathbb{E} \left[\sum_{t=1}^L U_d(b_{a'_t}(\mathbf{p}_t), \mathbf{p}_t) \right] \\ \leq c_2 L^{2/3} n |C| \log^{1/3}(n|C|). \end{aligned} \quad (37)$$

Proof. (a) The bound of Balcan et al. [2, Theorem 6.1] already holds at every prefix of a run tuned to a horizon H . Their algorithm divides $[1, H]$ into $Z = n(H^2 \log(n|C|))^{1/3}$ estimation blocks of H/Z rounds each, with Z a function of H , n and $|C|$ only. Dispose first of the degenerate range $Z \geq H$, in which there are not enough rounds to form the blocks at all: that range is $H \leq n^3 \log(n|C|)$, and there the trivial bound of two per round gives regret at most $2H = 2H^{1/3} \cdot H^{2/3} \leq 2n \log^{1/3}(n|C|) H^{2/3}$, which is below the claim for any $c_0 \geq 2$. So assume $Z \leq H$. The chain of inequalities in their proof of Lemma 6.2 bounds the algorithm's loss over the first τ_0 blocks by the loss of the best fixed action over those same blocks plus

$$\frac{H}{Z} R_{\tau_0, \kappa} + \tau_0 |C|,$$

where $\kappa = n|C|$ is the range of the loss estimator [2, Lemma 6.4], $R_{\tau_0, \kappa}$ is the regret of the full-information sub-routine after τ_0 estimated loss vectors, and the term $\tau_0 |C|$ pays for the $|C|$ exploration rounds of each block. Both summands are non-decreasing in τ_0 and are maximized at $\tau_0 = Z$; and the sub-routine of Proposition 3 is anytime, so $R_{\tau_0, \kappa} = O(\sqrt{\tau_0 \kappa \log |\mathcal{E}(C; \epsilon)|})$ holds for every τ_0 and not only for $\tau_0 = Z$. Reading the chain at an arbitrary $\tau_0 \leq Z$ therefore reproduces their bound with H in place of T . A prefix that stops

inside a block contributes at most two per round for the at most H/Z rounds of that block, and $H/Z \leq H^{2/3}n|C|\log^{1/3}(n|C|)$, so this is absorbed. The discretization error incurred by optimizing over $\mathcal{E}(C; \epsilon)$ instead of $\mathcal{P}$ is $2\epsilon L \leq 2$ by Lemma 1 and $\epsilon \leq 1/T$, and is absorbed as well; note that ϵ enters the size bound of Lemma 2 not at all, so taking it smaller than Balcan et al. [2] do is free. Hence there is an absolute constant c_0 such that a run tuned to H satisfies Equation (37) with $L^{2/3}$ replaced by $H^{2/3}$, simultaneously for every $L \leq H$.

(b) *Doubling*. Write $\phi = n|C|\log^{1/3}(n|C|)$ for the catalogue-dependent factor in Equation (37), and let $\mathcal{A}(C)$ run in stages $s = 0, 1, 2, \dots$, where stage s occupies 2^s consecutive rounds and is a fresh instance tuned to the horizon $H_s = 2^s$. Fix L and let S be the stage containing round L . Stages $0, \dots, S-1$ occupy $2^S - 1$ rounds, all of them at most L , so $2^S \leq L$. For any $\mathbf{p} \in \mathcal{P}$ the sum over $[1, L]$ splits across stages, so the maximum over $\mathcal{P}$ of the sum is at most the sum over stages of the per-stage maxima; subtracting the algorithm's payoff stage by stage gives

$$R_{[1, L]} \leq \sum_{s=0}^S R^{(s)},$$

where $R^{(s)}$ denotes the regret of the instance of stage s over that part of stage s which lies in $[1, L]$. Each $R^{(s)}$ is bounded by (a) with $H = 2^s$ — the restriction of an oblivious sequence to a stage is oblivious and still takes values in C — so

$$R_{[1, L]} \leq c_0 \phi \sum_{s=0}^S 2^{2s/3} \leq \frac{2^{2/3}}{2^{2/3} - 1} c_0 \phi 2^{2S/3},$$

and $2^S \leq L$ gives Equation (37) with $c_2 = 2^{2/3}c_0/(2^{2/3} - 1) < 2.71 c_0$. $\square$

One caveat about the factor $n|C|$ in Equation (37) should be recorded, since Lemma 24 is the one place in this report where the proof of Balcan et al. [2, Theorem 6.1] is reopened rather than quoted. That factor is inherited, through their Lemma 6.2, from the form $R_{L', \kappa} \leq \sqrt{L' \kappa \log |\mathcal{M}|}$ in which Proposition 3 states the full-information guarantee for losses of range κ . The textbook guarantee for Polynomial Weights on losses in $[-\kappa, \kappa]$ is instead $O(\kappa \sqrt{L' \log |\mathcal{M}|})$, obtained by rescaling to $[0, 1]$, and the two agree only at $\kappa = 1$. In Theorem 6 the loss range is indeed $\kappa = 1$ and nothing is at stake; here it is $\kappa = n|C|$ [2, Lemma 6.4], and the difference matters. We state Lemma 24 and Theorem 28 with the factor exactly as Balcan et al. [2, Theorem 6.1] delivers it. A reader who prefers the rescaling bound should re-optimize their Lemma 6.2 with $R_{L', \kappa} = \kappa \sqrt{L' \log |\mathcal{M}|}$, which gives $O(L'^{2/3} |S|^{1/3} \kappa^{2/3} \log^{1/3} |\mathcal{M}|)$ and hence $n^{4/3} |C|^{4/3}$ in place of $n|C|$ in Equation (37), and $O(n^{4/3} K'^{5/3} T^{2/3} \log^{1/3}(nK') + 2K'B)$ in place of Equation (39). Nothing else in this section changes: the epoch decomposition, the Hölder step that produces the $K'^{1/3}$, and the additive $2K'B$ are all independent of which form is used.

10.3 Block structure

The analysis rests on a decomposition of the horizon that plays the role of Step 1 of the proof of Theorem 6, but at the granularity of blocks rather than rounds.

Algorithm 2 BLOCK-GROW-CATALOGUE (partial information, revelation length B).

Require: horizon T , revelation length B , game primitives, default $\mathbf{p}_{\text{def}} \in \mathcal{P}$

```

1:  $C \leftarrow \emptyset$ ;  $\epsilon \leftarrow 1/T$ ;  $\mathcal{A} \leftarrow \text{NULL}$ 
2: for  $t = 1, \dots, T$  do
3:   if  $C = \emptyset$  then
4:     play  $\mathbf{p}_t \leftarrow \mathbf{p}_{\text{def}}$ 
5:   else
6:     play the  $\mathbf{p}_t$  prescribed by  $\mathcal{A}$ 
7:   end if
8:   observe the attacked target  $i_t = b_{a_t}(\mathbf{p}_t)$ ; pass  $i_t$  to  $\mathcal{A}$ 
9:   if  $t = \nu B$  for some  $\nu$ , or  $t = T$  then ▷ checkpoint
10:    query the oracle for  $D_\nu$ 
11:    if  $D_\nu \neq C$  then ▷ discovery block
12:       $C \leftarrow D_\nu$ 
13:       $\mathcal{A} \leftarrow \mathcal{A}(C)$  ▷ restart; the state built during  $\mathcal{B}_\nu$  is discarded
14:    end if
15:  end if
16: end for

```

Definition 25 (discovery blocks and epochs). Call a block $\mathcal{B}_\nu$ a *discovery block* if $D_\nu \supsetneq D_{\nu-1}$ and a *clean block* otherwise, and let $\beta_1 < \beta_2 < \dots < \beta_m$ enumerate the discovery blocks. For $j \in [m]$ define the j -th *epoch* to be the set of rounds in the clean blocks that separate β_j from the next discovery block,

$$\text{Ep}_j = \bigcup_{\nu=\beta_j+1}^{\beta_{j+1}-1} \mathcal{B}_\nu, \quad L_j = |\text{Ep}_j|, \quad (38)$$

with the convention $\beta_{m+1} := N_B + 1$.

Lemma 26 (the decomposition). *Let $K' = |\{a_1, \dots, a_T\}| \leq K_{\max}$ be the number of distinct types that actually appear. Then*

1. $\beta_1 = 1$, and $m \leq K'$;
2. $[1, T] = (\bigcup_{j=1}^m \mathcal{B}_{\beta_j}) \sqcup (\bigcup_{j=1}^m \text{Ep}_j)$, and the discovery blocks contribute at most $mB \leq K'B$ rounds in total;
3. $\sum_{j=1}^m L_j \leq T$;
4. if $\mathcal{B}_\nu$ is a clean block then $a_t \in D_{\nu-1}$ for every $t \in \mathcal{B}_\nu$;
5. the whole decomposition is deterministic: $\beta_1, \dots, \beta_m$ and $\text{Ep}_1, \dots, \text{Ep}_m$ are functions of the sequence $\mathbf{a}$ alone, and do not depend on the defender's plays or on its internal randomization.

Proof. (i) $D_0 = \emptyset$ and $D_1 \neq \emptyset$ (round 1 is attacked by some type), so $\mathcal{B}_1$ is a discovery block. Each discovery block satisfies $|D_{\beta_j}| \geq |D_{\beta_{j-1}}| + 1$, so $|D_{\beta_m}| \geq m$; and $D_{\beta_m} \subseteq \{a_1, \dots, a_T\}$ gives $m \leq K'$.

(ii) By construction every block is either a discovery block or lies in exactly one Ep_j : the blocks strictly between β_j and β_{j+1} are clean by the definition of consecutive discovery blocks, and by (i) there are no blocks before $\beta_1 = 1$. Each block has at most B rounds, so the m discovery blocks contribute at most mB rounds.

(iii) Immediate from (ii), since the Ep_j are disjoint subsets of $[1, T]$.

(iv) Contrapositive. If $a_t \notin D_{\nu-1}$ for some $t \in \mathcal{B}_\nu$, then $a_t \in D_\nu \setminus D_{\nu-1}$ by Equation (36), so $D_\nu \supsetneq D_{\nu-1}$ and $\mathcal{B}_\nu$ is a discovery block.

(v) By Equation (36) the revealed set D_ν is a function of $a_1, \dots, a_{\min\{\nu B, T\}}$ only. Hence so is the predicate “ $\mathcal{B}_\nu$ is a discovery block”, and therefore so are $\beta_1, \dots, \beta_m$ and the epochs of Equation (38). $\square$

Part (iv) is the containment property that makes the whole analysis go through, and it deserves a sentence of emphasis. The defender cannot tell, *while a block is being played*, whether the type attacking at round t is one it knows about. What (iv) says is that it does not have to: a round whose type is outside the current catalogue can only occur inside a block that the very next checkpoint will flag as a discovery block. Contamination is therefore confined to blocks that are detected — one checkpoint late, but before any of the contaminated information has been allowed to influence a later epoch.

Lemma 27 (epochs are clean runs of a fixed, correct catalogue). *Fix $j \in [m]$ and let $C_j := D_{\beta_j}$ be the catalogue held by Algorithm 2 at the end of block β_j . Then*

1. $C_t = C_j$ for every $t \in \text{Ep}_j$;
2. $a_t \in C_j$ for every $t \in \text{Ep}_j$;
3. *throughout Ep_j the algorithm runs one uninterrupted instance $\mathcal{A}(C_j)$, started at the first round of Ep_j , and no estimate that this instance uses at a round of Ep_j is built on a round whose type lies outside C_j .*

Proof. (i) Algorithm 2 modifies C only at a checkpoint at which $D_\nu \neq C$, i.e. only at discovery blocks. The blocks $\beta_j + 1, \dots, \beta_{j+1} - 1$ are clean, so C is unchanged throughout Ep_j and equals its value $D_{\beta_j} = C_j$ at the end of β_j .

(ii) Let $t \in \text{Ep}_j$, say $t \in \mathcal{B}_\nu$ with $\beta_j < \nu < \beta_{j+1}$. By Lemma 26(iv), $a_t \in D_{\nu-1}$. Since no block strictly between β_j and ν is a discovery block, $D_{\nu-1} = D_{\beta_j} = C_j$.

(iii) By (i) the catalogue does not change during Ep_j , so no restart is triggered inside Ep_j ; and the instance was restarted at the checkpoint closing β_j , hence starts at the first round of Ep_j . Every round $t \in \text{Ep}_j$ has $a_t \in C_j$ by (ii), so every observation the instance consumes at a round of Ep_j is generated by a type in its own attacker set. The claim is restricted to rounds of Ep_j for a reason: the instance is *not* discarded when Ep_j ends. It keeps running through the next discovery block $\mathcal{B}_{\beta_{j+1}}$, and that block is precisely where, by Lemma 26(iv), a contaminated round can occur. What Algorithm 2 guarantees is that the instance is discarded at the checkpoint closing that block, so no contaminated observation is ever used in a later epoch — and, as Theorem 28 uses only the prefix Ep_j of the instance’s run, no contaminated observation enters the bound at all. $\square$

10.4 Main result

Theorem 28 (block revelation restores no-regret). *Let the defender play Algorithm 2 in the setting of Problem 23. For any oblivious adversarial sequence $\mathbf{a} \in K^T$ with $|K| \leq K_{\max}$, and with $K' \leq K_{\max}$ the number of distinct types appearing in $\mathbf{a}$, the expected regret satisfies*

$$\mathbb{E}[\text{Regret}(T)] \leq c n K'^{4/3} T^{2/3} \log^{1/3}(n K') + 2K' B \quad (39)$$

for an absolute constant c . In particular

$$\mathbb{E}[\text{Regret}(T)] = O\left(n K_{\max}^{4/3} T^{2/3} \log^{1/3}(n K_{\max}) + K_{\max} B\right). \quad (40)$$

Proof. Fix the oblivious sequence $\mathbf{a}$ and adopt the notation of Definition 25 and Lemma 26. The proof follows the skeleton of Section 6; the steps that differ are Steps 1, 3 and 6.

Step 1 (Block decomposition). By Lemma 26(ii) the horizon splits as

$$[1, T] = \underbrace{\bigcup_{j=1}^m \mathcal{B}_{\beta_j}}_{\text{discovery blocks}} \sqcup \underbrace{\bigcup_{j=1}^m \text{Ep}_j}_{\text{epochs}}, \quad m \leq K'.$$

Unlike Step 1 of Section 6, where the epochs were separated by *single* discovery rounds, here they are separated by whole blocks; this is the only structural change, and it is what produces the additive B in Equation (39).

Step 2 (Catalogue is constant and correct inside an epoch). This is Lemma 27(i)–(ii): $C_t = C_j$ and $a_t \in C_j$ for all $t \in \text{Ep}_j$. It is the exact analogue of Equation (9).

Step 3 (Within-epoch regret). By Lemma 27(iii), during Ep_j the defender runs one uninterrupted instance $\mathcal{A}(C_j)$ — the doubling wrapper of Lemma 24, not the horizon-dependent Balcan et al. [2, Algorithm 1] inside it — on an oblivious sequence of length L_j all of whose types lie in the known set C_j , with partial-information feedback and with no contaminated estimate. These are exactly the hypotheses of Lemma 24 with n targets and the catalogue C_j . Two points deserve care, and they are the two the lemma was stated to cover. First, the guarantee is proved for an *oblivious* sequence — Balcan et al. [2, Lemma 6.3] needs the attacker met at a uniformly random exploration round to be a uniformly random attacker of the block — and the sub-sequence $(a_t)_{t \in \text{Ep}_j}$ is indeed oblivious, because by Lemma 26(v) the epoch Ep_j is determined by $\mathbf{a}$ alone and not by the defender’s plays or by its randomization. Second, L_j is not known when the instance starts, and Lemma 24 is exactly the statement that this costs only a constant. Note also why Lemma 24 has to hold at *every* prefix and not merely at the end of a run: the instance keeps going into the discovery block $\mathcal{B}_{\beta_{j+1}}$, where it does see contaminated rounds, so the only guarantee we may use is the one at the prefix of length L_j , which stops before them. Hence, with c_2 the constant of Lemma 24,

$$\begin{aligned} \max_{\mathbf{p} \in \mathcal{P}} \sum_{t \in \text{Ep}_j} U_d(b_{a_t}(\mathbf{p}), \mathbf{p}) - \mathbb{E} \left[\sum_{t \in \text{Ep}_j} U_d(b_{a_t}(\mathbf{p}_t), \mathbf{p}_t) \right] \\ \leq c_2 L_j^{2/3} n |C_j| \log^{1/3}(n |C_j|). \end{aligned} \quad (41)$$

Note that, in contrast to Equation (10), the left-hand side already compares against the unrestricted maximum over $\mathcal{P}$: Balcan et al. [2, Theorem 6.1] states its bound against the best fixed strategy in hindsight, so the passage through $\mathcal{E}(C_j; \epsilon)$ and the associated $2\epsilon L_j$ term of Lemma 1 are internal to their result and need not be repeated here.

Step 4 (Pass from the epoch optimum to $\mathbf{p}^*$). Exactly as in Sub-step 4b of Section 6: the global benchmark $\mathbf{p}^*$ of Equation (3) is an element of $\mathcal{P}$ and is therefore a feasible competitor in the maximization on the left of Equation (41), so

$$\max_{\mathbf{p} \in \mathcal{P}} \sum_{t \in \text{Ep}_j} U_d(b_{a_t}(\mathbf{p}), \mathbf{p}) \geq \sum_{t \in \text{Ep}_j} U_d(b_{a_t}(\mathbf{p}^*), \mathbf{p}^*). \quad (42)$$

Combining Equations (41) and (42),

$$\begin{aligned} \sum_{t \in \text{Ep}_j} U_d(b_{a_t}(\mathbf{p}^*), \mathbf{p}^*) - \mathbb{E} \left[\sum_{t \in \text{Ep}_j} U_d(b_{a_t}(\mathbf{p}_t), \mathbf{p}_t) \right] \\ \leq c_2 L_j^{2/3} n |C_j| \log^{1/3}(n |C_j|). \end{aligned} \quad (43)$$

Step 5 (Discovery blocks). Let t lie in a discovery block. Since $U_d(i, \mathbf{p}) \in [-1, 1]$ for every target i and every $\mathbf{p} \in \mathcal{P}$, exactly as in Equation (15),

$$U_d(b_{a_t}(\mathbf{p}^*), \mathbf{p}^*) - U_d(b_{a_t}(\mathbf{p}_t), \mathbf{p}_t) \leq 2.$$

By Lemma 26(ii) there are at most $mB \leq K'B$ such rounds, so their total contribution to the regret is at most

$$2K'B. \quad (44)$$

This term is the entire price of batching the revelation, and it is charged at face value. We make no attempt to learn during a discovery block, and Theorem 32 shows that nothing is lost by not trying: a discovery block is exactly a stretch in which the defender faces a type from the infinite universe that it has not been told about, and on the instance of that theorem such a stretch costs $B - 2$ rounds however the defender plays.

Step 6 (Uniform bound on the catalogue). $|C_j| \leq |D_{\beta_m}| \leq K'$ for every j , so $|C_j| \log^{1/3}(n |C_j|) \leq K' \log^{1/3}(nK')$ uniformly in j , and this factor can be pulled out of the sum over epochs. This is the analogue of Step 7 of Section 6, and it is simpler here only because Balcan et al. [2, Theorem 6.1] has already absorbed $\log |\mathcal{E}(C_j; \epsilon)|$ into the displayed form of its bound.

Step 7 (Summing across epochs). Summing Equation (43) over $j \in [m]$ and adding Equation (44),

$$\mathbb{E}[\text{Regret}(T)] \leq c_2 n K' \log^{1/3}(nK') \sum_{j=1}^m L_j^{2/3} + 2K'B. \quad (45)$$

Step 8 (Hölder). The sum of m two-thirds powers is controlled by Hölder's inequality with exponents 3 and 3/2 applied to $(1, \dots, 1) \in \mathbb{R}^m$ and $(L_1^{2/3}, \dots, L_m^{2/3})$:

$$\sum_{j=1}^m L_j^{2/3} \leq m^{1/3} \left(\sum_{j=1}^m L_j \right)^{2/3} \leq K'^{1/3} T^{2/3}, \quad (46)$$

using $m \leq K'$ and $\sum_j L_j \leq T$ from Lemma 26. As with the Cauchy–Schwarz step Equation (19), equality requires all epochs to have the same length, which corresponds to an adversary that spreads its type reveals evenly across the horizon; this is where the extra $K'^{1/3}$ over Balcan et al. [2, Theorem 6.1] is generated.

Step 9 (Final substitution). Substituting Equation (46) into Equation (45) gives Equation (39) with $c = c_2$, and bounding $K' \leq K_{\max}$ gives Equation (40). $\square$

Remark 29 (the two prices, separated). Equation (39) differs from Balcan et al. [2, Theorem 6.1] with $k = K_{\max}$ in exactly two places, and the two are of different natures. The factor $K_{\max}^{1/3}$ is the *price of not knowing K* : it comes from Hölder over the at most $K_{\max}$ epochs in Step 8, and it is the partial-information counterpart of the factor $\sqrt{K_{\max}}$ that Cauchy–Schwarz produces in Step 9 of Section 6. The additive $2K_{\max}B$ is the *price of batching the revelation*, and by Theorem 32 it is unavoidable up to a factor $2B/(B-2)$, which is $2 + o(1)$ as B grows and is vacuous for $B \leq 2$. The exponent 2/3 is inherited unchanged from Balcan et al. [2, Theorem 6.1] and is not affected by the unknown K at all: block revelation supplies the identities of the types, but it supplies no information whatever about their frequencies, which is the quantity their barycentric-spanner estimator has to reconstruct. Any improvement of the exponent here would be an improvement of their Theorem 6.1.

It is worth seeing exactly which feature of Problem 23 buys the $K_{\max}^{1/3}$, and the cleanest way is to price a strictly stronger oracle. Suppose that at a single round $t = P$ the defender were told the whole of K , including the types that have not yet attacked. The first P rounds cost at most $2P$, and from $P+1$ onwards the defender is in the known- K partial-information setting of Balcan et al. [2, Theorem 6.1], so

$$\mathbb{E}[\text{Regret}(T)] \leq 2P + O((T-P)^{2/3} n K_{\max} \log^{1/3}(n K_{\max})),$$

with no $K_{\max}^{1/3}$ at all, and the reveal is free for every $P = O(T^{2/3} n K_{\max} \log^{1/3}(n K_{\max}))$. The extra $K_{\max}^{1/3}$ in Equation (39) is therefore not the price of an unknown K as such: it is the price of learning K *incrementally*, one type at a time, which is what forces the epoch decomposition and the Hölder step. An oracle that can name a type before it has ever attacked avoids it; the oracle of Problem 23, which only ever reports what has already happened, cannot.

Note also that Algorithm 2 never reads $K_{\max}$: the bound enters only the analysis, and Equation (39) is stated in terms of the number K' of types that actually appear. Equation (39) is therefore automatically instance-adaptive, in exactly the sense that Corollary 8 makes explicit for Algorithm 1, and the defender does not need to be told $K_{\max}$ at all — only T and B .

10.5 How coarse may the revelation be?

Corollary 30 (free batching, and the number of oracle calls). *The additive term of Equation (39) is dominated by the leading term as soon as $B \leq B^*$, where*

$$\begin{aligned} B^* &= \frac{1}{2} c n K_{\max}^{1/3} T^{2/3} \log^{1/3}(n K_{\max}) \\ &= \Theta(n K_{\max}^{1/3} T^{2/3} \log^{1/3}(n K_{\max})). \end{aligned} \quad (47)$$

For every such B , $\mathbb{E}[\text{Regret}(T)] = O(n K_{\max}^{4/3} T^{2/3} \log^{1/3}(n K_{\max}))$, i.e. batching is free. Equivalently, the defender need consult the oracle only

$$N_B = \lceil T/B^* \rceil = O\left(\frac{T^{1/3}}{n K_{\max}^{1/3} \log^{1/3}(n K_{\max})} \right) \quad (48)$$

times over the whole horizon, and this count is optimal: by Theorem 32, for every $N \geq K_{\max}$ no algorithm that receives N evenly spaced revelations can have worst-case regret R unless

$$N \geq \frac{K_{\max} T}{R + 2K_{\max}}, \quad (49)$$

which at $R = \Theta(nK_{\max}^{4/3}T^{2/3}\log^{1/3}(nK_{\max}))$ is $\Omega(T^{1/3}/(nK_{\max}^{1/3}\log^{1/3}(nK_{\max})))$, matching Equation (48) up to a constant. Batching is free up to $\Theta(T^{1/3})$ reports and not beyond. All of this is a genuine saving only when $B^* \leq T$, i.e. when $nK_{\max}^{1/3}\log^{1/3}(nK_{\max}) \leq c'T^{1/3}$ for $c' = 2/c$; outside that regime even a single revelation at $t = T$ is too coarse, and Corollary 31 already fails.

Proof. Setting $2K_{\max}B \leq cnK_{\max}^{4/3}T^{2/3}\log^{1/3}(nK_{\max})$ and solving for B gives Equation (47); the displayed regret bound is then Equation (40) with the second term absorbed into the first. Equation (48) is $\lceil T/B^* \rceil$. For Equation (49), N evenly spaced revelations over $[1, T]$ is Problem 23 with $B = T/N$; the hypothesis $T \geq K_{\max}B$ of Theorem 32 reads $N \geq K_{\max}$, and under it that theorem gives $R \geq K_{\max}(T/N - 2)$, which rearranges to Equation (49). (For $N < K_{\max}$ the same theorem applied with $K_{\max}$ replaced by N gives the stronger $R \geq T - 2N$, so such an N is never worth considering.) $\square$

Corollary 31 (regime of applicability). *The bound of Theorem 28 is sublinear in T if and only if*

$$nK_{\max}^{4/3}\log^{1/3}(nK_{\max}) = o(T^{1/3}) \quad \text{and} \quad K_{\max}B = o(T).$$

In particular, for $K_{\max} = O(1)$ and $B \leq B^$ the rate is $\tilde{O}(nT^{2/3})$.*

Proof. Divide Equation (40) by T and require each of the two terms to vanish. $\square$

10.6 The additive term is necessary

Theorem 32 shows that the $\Theta(K_{\max}B)$ term of Theorem 28 is not an artefact of the analysis. The construction is the gadget G_3 of Section 9.2 again, with the dyadic family of Equation (26) applied recursively: each new type hides inside the previous one, so that a single coverage vector is optimal against all of them, while each of them must be located afresh.

Theorem 32 (a matching lower bound in B). *Let G_3 and $\mathfrak{U}$ be as in Theorem 13. For every $B \geq 1$, every $K_{\max} \geq 1$ and every horizon $T \geq K_{\max}B$, there is a distribution over instances of Problem 23 with $|K| = K_{\max}$ on which every online algorithm — randomized or deterministic — suffers*

$$\mathbb{E}[\text{Regret}(T)] \geq K_{\max}(B - 2). \quad (50)$$

Proof. Put $M = 2^B$. Draw indices $k_1, \dots, k_{K_{\max}}$ independently and uniformly from $[M]$ and define nested intervals $J_0 \supset J_1 \supset \dots \supset J_{K_{\max}}$ by $J_0 = (0, 1]$ and

$$J_j = \left(\ell_{j-1} + \frac{k_j - 1}{M} |J_{j-1}|, \ell_{j-1} + \frac{k_j}{M} |J_{j-1}| \right], \quad (51)$$

where ℓ_{j-1} and $|J_{j-1}|$ are the left endpoint and the length of J_{j-1} ; that is, J_j is a uniformly random one of the M equal sub-intervals of J_{j-1} . Let $\alpha^{[j]} = \alpha_{r_1^{[j]}, r_2^{[j]}}$ be the type of Lemma 10

whose interval $(r_1^{[j]}, r_2^{[j]})$ is J_j , put $K = \{\alpha^{[1]}, \dots, \alpha^{[K_{\max}]}\}$, and let the sequence be $a_t = \alpha^{[j]}$ for every $t \in \mathcal{B}_j$ and $j \in [K_{\max}]$, and $a_t = \alpha^{[K_{\max}]}$ for every $t > K_{\max}B$. The sequence is fixed once $k_1, \dots, k_{K_{\max}}$ are drawn, so the adversary is oblivious. The $K_{\max}$ types are distinct, because $M = 2^B \geq 2$ makes each J_j a proper sub-interval of J_{j-1} , so $|K| = K_{\max}$ as required. As in Theorem 13 we lower bound the expected regret of an arbitrary deterministic algorithm and appeal to Yao's minimax principle.

Step 1 (the benchmark is 0). Let $\mu = \frac{1}{2}(r_1^{[K_{\max}]} + r_2^{[K_{\max}]})$ be the midpoint of $J_{K_{\max}}$ and put $\mathbf{p}^* = (\mu, 0, 1 - \mu) \in \mathcal{P}$. For every $j \in [K_{\max}]$ the nesting $J_{K_{\max}} \subseteq J_j$ gives $r_1^{[j]} \leq r_1^{[K_{\max}]} < r_2^{[K_{\max}]} \leq r_2^{[j]}$, hence $p_1^* = \mu > r_1^{[K_{\max}]} \geq r_1^{[j]}$ and $p_3^* = 1 - \mu > 1 - r_2^{[K_{\max}]} \geq 1 - r_2^{[j]}$, both strictly, so $b_{\alpha^{[j]}}(\mathbf{p}^*) = 2$ by Equation (25), under any tie-breaking rule. Thus $\mathbf{p}^*$ hits in every round and, as in Equation (33), the regret equals the expected number of missed rounds.

Step 2 (what the defender knows when block j starts). The checkpoint closing $\mathcal{B}_{j-1}$ reveals $D_{j-1} = \{\alpha^{[1]}, \dots, \alpha^{[j-1]}\}$ with utility vectors, hence J_{j-1} ; it reveals nothing about $\alpha^{[j]}$, which has not yet attacked. Moreover the entire transcript of blocks $\mathcal{B}_1, \dots, \mathcal{B}_{j-1}$ is a function of $k_1, \dots, k_{j-1}$ alone: by induction on the rounds, the defender's play at any round of those blocks is determined by the earlier observations, and each observation is determined by that play together with the interval of the type then attacking, which is one of $J_1, \dots, J_{j-1}$. Since k_j is drawn independently of $k_1, \dots, k_{j-1}$, it is therefore still uniform on $[M]$ conditionally on everything the defender has seen and been told before block j begins.

Step 3 (each block is a fresh comparison search). Fix j and condition on the transcript before $\mathcal{B}_j$. Under that conditioning the defender's play at each round of $\mathcal{B}_j$ is a deterministic function of the observations made earlier *within* $\mathcal{B}_j$, so the B rounds of the block form a protocol of the kind covered by Lemma 12, with k_j uniform on $[M]$ by Step 2. During $\mathcal{B}_j$ the only attacker is $\alpha^{[j]}$, whose interval is one of the M pairwise disjoint sub-intervals of J_{j-1} indexed by k_j ; applying Lemma 11 to that family, for every $\mathbf{p} \in \mathcal{P}$ at most one candidate index yields $b(\mathbf{p}) = 2$, which is the hypothesis Equation (28). The conditioning has already absorbed everything the oracle said before $\mathcal{B}_j$, and it says nothing during $\mathcal{B}_j$, so the budget is again $b = 0$. Lemma 12 with $M = 2^B$ and horizon B gives

$$\mathbb{E}[\#\{\text{misses in } \mathcal{B}_j\}] \geq B - \frac{2^{B+1} - B - 2}{2^B} > B - 2.$$

Step 4 (summing and Yao). Summing over $j \in [K_{\max}]$ and using Step 1, $\mathbb{E}[\text{Regret}(T)] \geq K_{\max}(B - 2)$ for every deterministic algorithm. Averaging over the internal randomness of a randomized algorithm as in Step 4 of the proof of Theorem 13 gives Equation (50). $\square$

Remark 33 (the additive term is tight). Theorems 28 and 32 bracket the dependence on the revelation granularity: the additive cost of batching is between $K_{\max}(B - 2)$ and $2K_{\max}B$, so it is $\Theta(K_{\max}B)$ for $B \geq 3$, the two bounds differing by the

factor $2B/(B-2) — 6$ at $B = 3$, 4 at $B = 4$, and $2 + o(1)$ thereafter. For $B \leq 2$ the lower bound is vacuous and only the upper bound $2K_{\max}B$ survives. Two special cases are worth recording. Taking $K_{\max} = 1$ and $B = T$ recovers $\mathbb{E}[\text{Regret}(T)] \geq T - 2$, i.e. Theorem 13: the impossibility is the $B \rightarrow T$ limit of Theorem 32. Taking $B = 1$ makes the additive term $2K_{\max}$ and the bound of Theorem 28 becomes $O(nK_{\max}^{4/3}T^{2/3} \log^{1/3}(nK_{\max}) + K_{\max})$, which is the partial-information analogue of Theorem 6 and, to our knowledge, the first sublinear bound for an unknown attacker set under partial-information feedback.

10.7 Scheduled versus chosen revelation times

Problem 23 fixes the checkpoints before play, which is what a threat-intelligence subscription looks like: reports arrive on a calendar, not on demand. It is worth recording how much of the $\Theta(K_{\max}B)$ term is due to that calendar rather than to the number of reports, because the two are easy to confuse. Corollary 30 counts reports; Theorem 32 charges for their timing. The next proposition separates the two as sharply as possible: on the very instances of Theorem 32, the same number of reports costs $T - 2K_{\max}$ when their times are fixed in advance and only $K_{\max}$ when the defender may pick them.

Proposition 34 (choosing the revelation times is exponentially cheaper). *Consider the variant of Problem 23 in which the defender is given a budget of N queries instead of a schedule: at the end of any round it may spend one query and receive, as in Equation (36), the set of types that have attacked so far together with their utility vectors. Let the instance be played in G_3 and let its types $\alpha_{r_1^{[1]}, r_2^{[1]}}, \dots, \alpha_{r_1^{[K']}, r_2^{[K']}}$ have nested threshold intervals, i.e. the intervals $(r_1^{[j]}, r_2^{[j]})$ form a chain under inclusion; this is the case for every instance constructed in Theorem 32. Then the defender that*

1. *plays the midpoint of the smallest interval among the types revealed so far (and any fixed $\mathbf{p}_{\text{def}}$ before the first revelation), and*
2. *spends a query at the end of every round in which target 1 or target 3 was attacked,*

spends at most K' queries — so a budget of $N = K_{\max}$ always suffices — and suffers $\text{Regret}(T) \leq K' \leq K_{\max}$, for every horizon T and every order in which the types appear. By contrast, Theorem 32 with $B = T/K_{\max}$ says that $N = K_{\max}$ revelations placed on a fixed schedule leave regret at least $T - 2K_{\max}$.

Proof. Let D be the set of types revealed so far and, when $D \neq \emptyset$, let $J(D)$ be the smallest of their intervals, which exists because the intervals form a chain; write μ for its midpoint and $\mathbf{p} = (\mu, 0, 1 - \mu)$ for the played coverage vector. For every $\alpha = \alpha_{r_1, r_2} \in D$ we have $J(D) \subseteq (r_1, r_2]$, so $r_1 < \mu < r_2$, and Equation (25) gives $b_\alpha(\mathbf{p}) = 2$: the play hits every revealed type, under any tie-breaking rule, since target 2 is the strict maximizer. Consequently a round in which target 1 or 3 is attacked — the only rounds in which the defender queries, and by Equation (21) the only rounds that cost anything — is a round whose attacker is *not* in D . The query issued at the end of such a round returns a set containing that attacker, so it enlarges D by at least one type. Since $|D| \leq K'$ throughout,

there are at most K' such rounds in the whole horizon; the same count bounds the number of queries.

The benchmark is 0: taking $\mathbf{p}^*$ to be the midpoint of the smallest interval of *all* K' types makes every round a hit by the same computation, and $U_d \leq 0$ everywhere in G_3 , so no $\mathbf{p}$ does better. By Equation (33) the regret is the number of missed rounds, which we have just bounded by K' . $\square$

Remark 35 (why this does not extend for free). Proposition 34 rests on a feature of G_3 that a general security game does not have: there, a missed round is a *certificate* that the catalogue is stale, so the defender knows exactly when to spend a query. In a general SSG a type outside the catalogue may best-respond at $\mathbf{p}_t$ exactly as the catalogue predicts and still differ elsewhere in $\mathcal{P}$, corrupting the frequency estimates of Algorithm 2 without ever announcing itself; and conversely a round can look anomalous for reasons that have nothing to do with a new type, because Algorithm 2 deliberately plays exploration strategies. Whether a defender with N self-timed queries can beat the $\Theta(K_{\max}T/N)$ of Theorem 32 in a general repeated SSG is open; Proposition 34 shows only that the lower bound is a statement about schedules and not about report counts.

11 Tractable Experiments

The experiments of Section 7 were run at $n = 3$ targets. The reason was not the online algorithm but the expert set: Algorithm 1 needs $\mathcal{E}(C; \epsilon)$ explicitly, and the implementation builds it by enumerating every $(n-1)$ -subset of the $H(n, |C|) = n + |C| \binom{n}{2}$ hyperplanes that define the partition. That is $\binom{H(n, |C|)}{n-1}$ linear solves, which is 1.5×10^5 at $n = 5$, $|C| = 4$ but 7.3×10^{14} at $n = 10$ and 5.9×10^{37} at $n = 20$.

This section removes that obstacle and reruns the experiments at $n = 10$ and $n = 20$, for Algorithm 1 and for Algorithm 2. The idea is that neither algorithm ever needs the set $\mathcal{E}(C; \epsilon)$ itself; each needs only to *optimize a linear function over it*, and that can be done with one call to a Bayesian-Stackelberg solver.

11.1 An oracle-efficient implementation

For a catalogue C and $\mathbf{p} \in \mathcal{P}$ define the *lift*

$$y_{\alpha, i}(\mathbf{p}) = \mathbb{I}\{b_\alpha(\mathbf{p}) = i\}, \quad w_{\alpha, i}(\mathbf{p}) = y_{\alpha, i}(\mathbf{p}) p_i \quad (52)$$

for $(\alpha, i) \in C \times N$, and let $\mathcal{S}(C) = \{(y(\mathbf{p}), w(\mathbf{p})) : \mathbf{p} \in \mathcal{P}\} \subseteq \mathbb{R}^{2|C|n}$.

Lemma 36 (both algorithms are linear over the lift). *Assume the types in C have pairwise distinct best-response maps, so that the set $\mathcal{W} = \{\mathbb{I}_{(\sigma=i)}\}$ of Balcan et al. [2, Section 6.3] has rank $|C|$ and its barycentric spanner Λ is invertible. Let $\omega \in \mathbb{R}^C$ be any vector of weights on the catalogue. Then*

$$\sum_{\alpha \in C} \omega_\alpha U_d(b_\alpha(\mathbf{p}), \mathbf{p}) = \sum_{\alpha \in C} \sum_{i \in N} \left[\omega_\alpha u_d^u(i) y_{\alpha, i}(\mathbf{p}) + \omega_\alpha (u_d^c(i) - u_d^u(i)) w_{\alpha, i}(\mathbf{p}) \right], \quad (53)$$

which is a linear function of the lift. Moreover:

1. the loss used by Algorithm 1 at round t is $\ell_t(\mathbf{p}) = -U_d(b_{a_t}(\mathbf{p}), \mathbf{p})$, i.e. Equation (53) with $\omega = -\mathbb{I}_{\{a_t\}}$;
2. the estimator $\hat{c}_\tau$ of Balcan et al. [2, Eq. 2 and Lemma 6.4], used inside Algorithm 2, is Equation (53) with $\omega = g_\tau$ — and the loss it induces is its negation, i.e. $\omega = -g_\tau$ — where $g_\tau = \Lambda^{-1} \hat{p}_\tau$, Λ is the matrix whose rows are the barycentric spanner $b_1, \dots, b_{|C|}$ of Balcan et al. [2, Section 6.3] and $\hat{p}_\tau \in \{0, 1\}^{|C|}$ collects that block's exploration outcomes. The vector g_τ is an unbiased estimator of the per-round type frequencies in the block.

Proof. For each α exactly one of $y_{\alpha,1}, \dots, y_{\alpha,n}$ equals 1, so $U_d(b_\alpha(\mathbf{p}), \mathbf{p}) = \sum_i y_{\alpha,i}(\mathbf{p}) U_d(i, \mathbf{p})$. By Equation (1),

$$\begin{aligned} y_{\alpha,i} U_d(i, \mathbf{p}) &= u_d^c(i) y_{\alpha,i} p_i + u_d^u(i) y_{\alpha,i} (1 - p_i) \\ &= u_d^u(i) y_{\alpha,i} + (u_d^c(i) - u_d^u(i)) y_{\alpha,i}, \end{aligned}$$

and summing against ω gives Equation (53). Item (i) is the special case $\omega = -\mathbb{I}_{\{a_t\}}$.

For (ii), recall that Balcan et al. [2, Eq. 2] sets $\hat{c}_\tau(\mathbf{p}) = \sum_i \sum_j \lambda_j(\mathbb{I}_{(\sigma=i)}) \hat{p}_\tau(b_j) U_d(i, \mathbf{p})$, where σ is the response profile of $\mathbf{p}$ and $\lambda(w)$ is the (unique, since Λ is invertible) vector of coefficients with $\sum_j \lambda_j(w) b_j = w$, i.e. $\lambda(w)^\top \Lambda = w^\top$. Hence $\sum_j \lambda_j(w) \hat{p}_\tau(b_j) = \lambda(w) \cdot \hat{p}_\tau = w^\top \Lambda^{-1} \hat{p}_\tau = w \cdot g_\tau$. The α -th coordinate of $\mathbb{I}_{(\sigma=i)}$ is exactly $y_{\alpha,i}(\mathbf{p})$, so

$$\begin{aligned} \hat{c}_\tau(\mathbf{p}) &= \sum_i \left(\sum_\alpha g_{\tau,\alpha} y_{\alpha,i}(\mathbf{p}) \right) U_d(i, \mathbf{p}) \\ &= \sum_\alpha g_{\tau,\alpha} U_d(b_\alpha(\mathbf{p}), \mathbf{p}), \end{aligned}$$

which is Equation (53) with $\omega = g_\tau$. Finally, Balcan et al. [2, Lemma 6.3] gives $\mathbb{E}[\hat{p}_\tau(b_j)] = f_\tau(b_j)/|B_\tau|$ and $f_\tau(w) = w \cdot m_\tau$ with m_τ the vector of type counts in the block, so $\mathbb{E}[\hat{p}_\tau] = \Lambda m_\tau / |B_\tau|$ and $\mathbb{E}[g_\tau] = m_\tau / |B_\tau|$. $\square$

Lemma 36 says that both algorithms are instances of online linear optimization over $\mathcal{S}(C)$, whose single primitive is

$$\text{ORACLE}(A, \Delta) = \arg \max_{\mathbf{p} \in \mathcal{P}} \sum_{\alpha,i} (A_{\alpha,i} y_{\alpha,i}(\mathbf{p}) + \Delta_{\alpha,i} w_{\alpha,i}(\mathbf{p})). \quad (54)$$

We therefore replace the Polynomial-Weights sub-routine of Proposition 3 by Follow the Perturbed Leader over $\mathcal{S}(C)$, which calls Equation (54) once per round — and no times at all on the exploration rounds of Algorithm 2, which play a fixed spanner strategy. This is a change of implementation and not of algorithm: Balcan et al. [2, Section 3.1] explicitly allow any black box satisfying their Proposition 1, and name Follow the Lazy Leader of Kalai and Vempala [12] as an admissible choice. We run it with the perturbation schedule of Kalai and Vempala [12], in its anytime form because the segment lengths are not known in advance. The three problem constants that schedule needs take, on $\mathcal{S}(C)$, the values $\|Y - Y'\|_1 \leq 4|C|$ for the ℓ_1 -diameter of the decision set, $|Y \cdot s| \leq 1$ for the per-round loss, and $\|s\|_1 \leq 3n$ for the ℓ_1 -norm of a one-round loss vector, the last because $\sum_i (|u_d^u(i)| + |u_d^c(i) - u_d^u(i)|) \leq 3n$.

Grouping Equation (54) by the response profile $\sigma : C \rightarrow N$ makes the objective linear on each cell $\mathcal{P}_\sigma(C)$ of Equation (6), so Equation (54) is a maximum of $n^{|C|}$ linear programs in

n variables — polynomial in n for a fixed catalogue, and the natural extension to a set of types of the multiple-LP method of Conitzer and Sandholm [6]. In the experiments we instead solve one mixed-integer program with $n|C|$ binaries in the style of the Bayesian-Stackelberg MILP of Paruchuri et al. [18], using the profile enumeration as a correctness reference. We do not claim polynomial time in $|C|$: the Bayesian Stackelberg problem is NP-hard [6]. The claim is polynomial time in n , which is the parameter that made $\mathcal{E}(C; \epsilon)$ unusable. Both back-ends solve over cells shrunk by a strictness margin $\gamma = 10^{-6}$, so that the returned $\mathbf{p}$ realizes its profile under any tie-breaking rule; this plays the role of the ϵ -approximation in Lemma 1. The rank hypothesis of Lemma 36 is not an extra assumption on the game but a normalisation of the catalogue: two types with the same best-response map on all of $\mathcal{P}$ are indistinguishable to the defender's payoff, to the benchmark Equation (3) and to the feedback, so Algorithm 2 keeps one representative of each best-response map in C and discards the duplicates. This can only shrink the catalogue, so $|C_j| \leq K'$ and every bound of Section 10 is unaffected; without it $\mathcal{W}$ can have rank below $|C|$ and Λ^{-1} in Lemma 36(ii) need not exist. The implementation does not de-duplicate but detects the failure: if the sampled pool admits no full-rank spanner it records a rank failure for that catalogue and skips exploration until the catalogue next changes. Over the runs behind Figure 4 no rank failure occurred, which is what one expects of independently drawn utility vectors. As in Section 7, the experiments take $\mathcal{P} = \{\mathbf{p} \geq 0 : \sum_i p_i = 1\}$ with one resource; the algorithm and the benchmark $\mathbf{p}^*$ optimize over the same set, so the regret they define is the one of Equation (4). Every non-asymptotic claim of Sections 9 and 10, together with the oracle and estimator claims of this subsection, is re-checked by `code/verify_theory.py`, which prints a pass or fail for each; the measured quantities of Sections 11.2 to 11.5 — fitted exponents, envelope constants and timings — are printed by the plotting and benchmarking scripts that produce the corresponding figure or table, and are reproduced by rerunning those.

One further gap between Algorithm 2 and its implementation should be stated. The set $\mathcal{W} = \{\mathbb{I}_{(\sigma=i)}\}$ of Balcan et al. [2, Section 6.3] has one element per pair (cell, target) and we do not enumerate it; instead we sample coverage vectors, read off the vectors $\mathbb{I}_{(\sigma=i)}$ they realize together with their witnesses $(\mathbf{p}_b, i_b)$, and run the greedy determinant maximization of Awerbuch and Kleinberg [1] on that pool. The consequence is confined to the *range* of the estimator: a spanner of a sub-pool need not give coefficients $\lambda \in [-1, 1]$ for every $w \in \mathcal{W}$, so the bound $\hat{c}_\tau \in [-n|C|, n|C|]$ of Balcan et al. [2, Lemma 6.4] is not guaranteed by the implementation. Unbiasedness — Lemma 36(ii), which is what the estimator's correctness rests on — needs only that Λ be invertible, and holds exactly; it is the check reported in the previous paragraph. For the same reason the implementation does not scale the Follow-the-Perturbed-Leader perturbation by that worst-case range: inside Algorithm 2 it uses the schedule $\epsilon_t = \sqrt{4|C|/(3nt)}$ listed above, which is what the range of the estimates we actually compute warrants, rather than the range $n|C|$ that Balcan et al. [2, Lemma 6.4] proves in the worst case. This is a tuning choice and not a claim: Theorem 28 rests on Lemma 24 and on their Lemma 6.4, not on the constants used here, and

n	$ C $	$\binom{H}{n-1}$	build $\mathcal{E}$	$n^{ C }$ LPs	MILP
3	4	1.1×10^2	6 ms ($ \mathcal{E} = 61$)	32 ms	16 ms
5	4	1.5×10^5	307 s ($ \mathcal{E} = 13118$)	230 ms	64 ms
10	4	7.3×10^{14}	infeasible	3.1 s	133 ms
20	4	5.9×10^{37}	infeasible	61 s	0.88 s
30	4	1.4×10^{63}	infeasible	infeasible	0.89 s
40	4	1.2×10^{90}	infeasible	infeasible	1.56 s

Table 3: Cost of one step, with $H = H(n, |C|) = n + |C| \binom{n}{2}$. Column 3 is the number of $(n-1)$ -subsets that the exhaustive construction of $\mathcal{E}(C; \epsilon)$ must solve. Columns 5 and 6 are the two back-ends for Equation (54); wherever both ran they returned the same optimum to 2×10^{-6} . Median of 10 calls on an otherwise idle machine; `code/bench_oracle.py`. The MILP is not monotone in n because it is an exact branch-and-bound solve, not a fixed amount of work.

what Figure 4 sets out to test is whether the predicted *exponent* appears in a run. Over the 60 random instances with $n \in \{3, 4, 5\}$ and $|C| \in \{1, 2, 3\}$ that `verify_theory.py` runs, the two back-ends and a brute-force maximization over the enumerated $\mathcal{E}(C; \epsilon)$ agree to within 1.4×10^{-6} , and a longer run of 150 instances does not push that figure up. We also checked Lemma 36(ii) directly: over the 4×10^4 replications of a single estimation block at $n = 6$, $|C| = 3$ that the same script runs, the mean of $|B_\tau| g_\tau$ reproduced the true type counts of the block to within 0.05 out of 40, comfortably inside the three-standard-error band of 1.1 that Monte-Carlo noise allows; a separate run at 2×10^5 replications gave a largest error of 0.09 against a standard error of 0.16, which is what an unbiased estimator should do.

11.2 Tractability

Table 3 times the three routines. Building $\mathcal{E}(C; \epsilon)$ already takes 307 seconds at $n = 5$, $|C| = 4$ and is out of reach beyond that, whereas a single oracle call costs 133 ms at $n = 10$, 0.9 s at $n = 20$ and 1.6 s at $n = 40$. Since neither algorithm makes more than one oracle call per round — and Algorithm 2 makes none at all on its exploration rounds — this is what moves the experiments from $n = 3$ to $n = 20$.

11.3 Full information at $n = 10$ and $n = 20$

Figure 3 reruns the two experiments of Section 7 with the oracle implementation, at $n = 10$ and $n = 20$ instead of $n = 3$, with $K_{\max} = 4$ and $T = 1500$. Panels (a) and (b) reproduce Figure 1 under the staggered-uniform adversary: the per-round regret spikes shortly after each of the four phase boundaries — a new type becomes available at a boundary and is first drawn a few rounds later, which is the discovery round — and then decays inside the epoch that follows, exactly the structure that Steps 1–6 of the proof of Theorem 6 describe. Panels (c) and (d) reproduce Figure 2 under the uniform-random adversary: the cumulative regret stays below the $C\sqrt{t}$ envelope of Theorem 6 throughout, with $C = 1.27$ at $n = 10$ and $C = 1.71$ at $n = 20$, and it is 8.0 and 11.6 times smaller than the uniform baseline.

We keep the two adversaries separate for the same reason Section 7 does, and it is worth spelling out. The staggered-

uniform adversary is non-stationary by design — the type mixture changes at every phase boundary — so the best *fixed* strategy in hindsight of Equation (3) is a weak comparator, and an algorithm that tracks the current phase can and does beat it. At $n = 10$ the cumulative regret against $\mathbf{p}^*$ came out negative on one of the four seeds (-35.3), and the four-seed mean was -2.6 . That says nothing against Theorem 6, which is an upper bound, but it makes a cumulative-regret plot on that adversary uninformative. The uniform-random adversary is stationary, so $\mathbf{p}^*$ is the right comparator, and panels (c) and (d) use it. The benchmark $\mathbf{p}^*$ is itself computed with one oracle call, so even *evaluating* the regret at $n = 20$ was out of reach before Section 11.1.

11.4 Block revelation at $n = 10$ and $n = 20$

Figure 4 runs Algorithm 2 under genuine partial-information feedback: the simulator passes the algorithm the attacked target only, and at the end of each revelation block calls it with the *set* of types seen so far and nothing else. The set is handed over as a list for convenience, but the algorithm reads it only through membership tests, so neither the order in which the types first attacked nor any frequency information can leak through it. Panels (a) and (b) show cumulative regret against the $Ct^{2/3}$ envelope of Theorem 28 under the staggered-uniform adversary, which is the interesting one here because it spreads the discovery blocks across the horizon and so produces the $\Theta(K_{\max})$ epochs that Step 8 of the proof pays for; the envelope constants are $C = 1.72$ at $n = 10$ and $C = 1.25$ at $n = 20$.

Panel (c) sweeps the revelation length B over six doublings at fixed T . The measured growth is linear in B , as Theorems 28 and 32 predict — a least-squares fit over $B \in [4, 256]$ has slope 0.47 — but that slope is far below the worst-case $2K' = 8$, and the panel draws both. The gap is expected rather than a discrepancy: the worst case is attained only by the nested adversary of Theorem 32, which Figure 5(d) exhibits separately, and on a random instance most of the rounds of a discovery block are not in fact wasted, because a strategy tuned to the types already in the catalogue is usually not terrible against a new one.

Panel (d) compares the two feedback models as T grows, on the stationary uniform-random adversary for the reason given in Section 11.3. Over $T \in \{250, 500, 1000, 2000\}$ the block-revelation regret grows with fitted exponent 0.664, which is $2/3$ to within the resolution of the experiment and is exactly the exponent of Theorem 28; the full-information regret grows with exponent 0.317, comfortably inside the $1/2$ of Theorem 6. The asymmetry is not an accident of the instances. Algorithm 2 pays a floor of $|C|$ exploration rounds in every estimation block whatever the sequence does, so its $T^{2/3}$ is visible even on an easy sequence, whereas Algorithm 1 has no such floor and, as already observed in Section 7, runs well inside its worst case against a stochastic adversary.

One implementation choice is worth stating, because it is where asymptotics and simulation disagree. Balcan et al. [2, Algorithm 1] uses $Z = n(T^2 \log(n|C|))^{1/3}$ estimation blocks, i.e. blocks of $T/Z = T^{1/3}/(n \log^{1/3}(n|C|))$ rounds, which at $n = 10$, $|C| = 4$ and $T = 2000$ is less than one round — fewer, certainly, than the $|C| = 4$ exploration rounds a block must

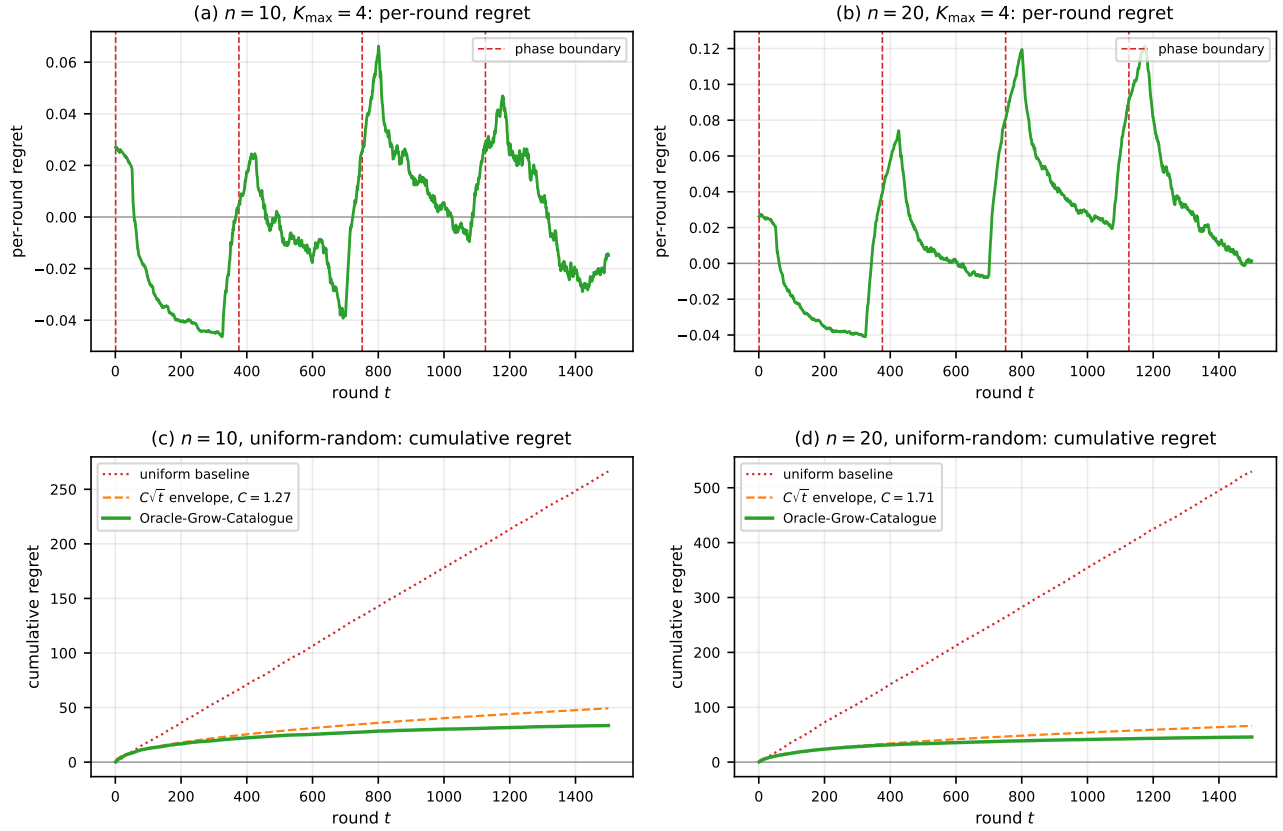


Figure 3: Algorithm 1 with the oracle implementation of Section 11.1, $K_{\max} = 4$, $T = 1500$, mean over 4 seeds. (a), (b) per-round regret under the staggered-uniform adversary, rolling window 101; red dashed lines mark the phase boundaries, at which a new type becomes available. (c), (d) cumulative regret under the uniform-random adversary, against the $C\sqrt{t}$ envelope of Theorem 6 and the uniform baseline.

hold, since the tuning of Balcan et al. [2, Theorem 6.1] is asymptotic. We therefore use blocks of $\max\{|C| + 1, \lceil T^{1/3} \rceil\}$ rounds, which has the same order in T . At $T = 2000$ and $K_{\max} = 4$ that is 13 rounds, of which 4 are spent exploring, so roughly 31% of all rounds are exploration; this is the main reason the measured regret in panels (a) and (b) is an order of magnitude above the full-information regret of Figure 3 at the same horizon, and it is a constant-factor effect, not a change of exponent.

11.5 The lower bounds, in simulation

Figure 5 plays the gadget G_3 of Section 9.2 for real: attacker types are built from the utility vectors of Equation (22), best responses are computed from those utilities, and the defender’s payoff is read off the attacked target. The defender is the optimal one for this game, namely the midpoint version-space search of Theorem 16(ii), and the expectation over the hidden type is computed exactly, by averaging over all M candidates rather than by sampling.

Panel (a) is Theorem 13: with $M = 2^T$ the measured expected regret equals $T - 2 + (T + 2)2^{-T}$ for every $T \leq 14$, to machine precision, confirming both the lower bound and the claim in Remark 14 that it is exactly attained.

Panel (b) is Theorem 16. The universe is fixed at $M = 2^{10}$ and the defender is helped by an oracle of budget $b \in \{0, 2, 4, 6\}$ that names the block of $M/2^b$ indices containing the true type, exactly as in the proof of Theorem 16(ii); everything else is the real game. Each curve saturates, and it saturates at $\log_2 M - b - 2$, which is the lower bound of Theorem 16(i): the measured values at $T = 14$ are 8.01, 6.04, 4.13 and 2.38 against predictions 8, 6, 4 and 2. The hollow markers are a separate run with budget 0 on the smaller universe $\mathfrak{U}_{M/2^b}$, and they land on the corresponding curve to machine precision — the regret depends on M and b only through $\log_2 M - b$, which is what makes Theorem 16 an accounting identity rather than two unrelated bounds. The case $b = 0$ also says that the linear growth in panel (a) is caused by the universe being infinite and by nothing else.

Panel (c) is Proposition 21, and it is the one panel that shows a T -dependence rather than a saturation. With $|K| = 2$ on the two-element universe, follow the leader has measured regret 1.57, 6.35 and 25.60 at $T = 16, 256$ and 4096, against the exact optimum $\frac{1}{2}\mathbb{E}|N_1 - N_2| = 1.571, 6.377, 25.531$ and the floor $\frac{1}{2}\sqrt{\lceil T/2 \rceil} = 1.41, 5.66, 22.63$. The dotted line is the same universe with $|K| = 1$, where the regret is 0.5 at every horizon. The two curves are the phase transition of Remark 22.

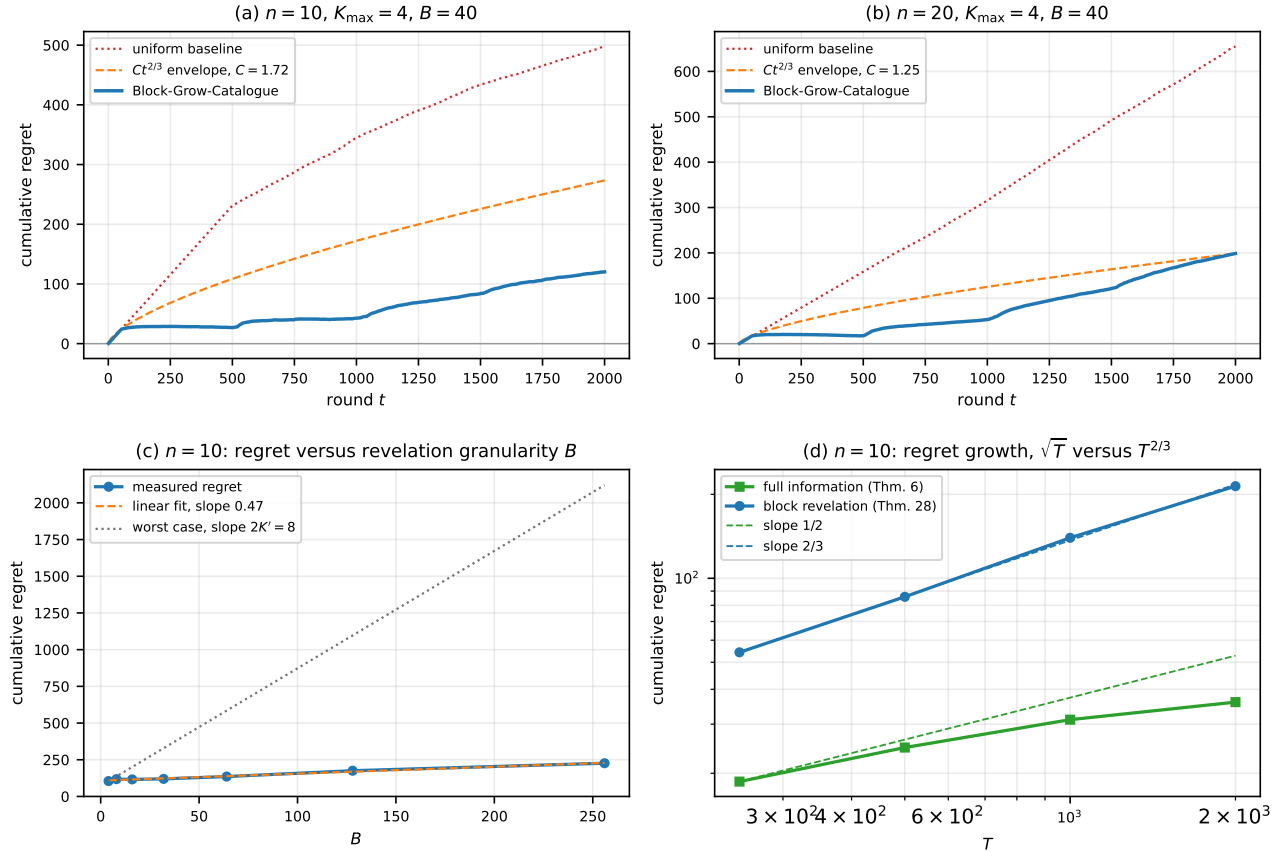


Figure 4: Algorithm 2 under partial-information feedback with block revelation, $K_{\max} = 4$, $T = 2000$, mean over 4 seeds. (a)–(c) staggered-uniform adversary; (d) uniform-random adversary. (a), (b) cumulative regret against the $Ct^{2/3}$ envelope of Theorem 28. (c) final regret as a function of the revelation length B , against a least-squares fit and against the worst-case slope $2K'$ of Theorem 28. (d) full information (Theorem 6) against block revelation (Theorem 28) as T grows; fitted exponents 0.317 and 0.664.

Panel (d) is Theorem 32: on the nested-interval instance the measured regret tracks $|K|(B-2)$ across $|K| \in \{1, 2, 3, 4\}$ and $B \in \{4, 8, 16, 32\}$, so the additive $\Theta(K_{\max}B)$ of Theorem 28 is real and not an artefact of the analysis.

Panel (e) is Proposition 34, on the same instances as panel (d) with $|K| = 4$ and $T = |K|B$, so that the two schemes are compared at the same report budget $N = |K| = 4$. Scheduling the four reports on a fixed grid costs $|K|(B-2)$, which is 120 rounds at $B = 32$; letting the defender choose their times costs exactly 4, and the measured query count is 4.00, so the defender does not even exceed its budget. The measured adaptive regret is 3.72, 3.98, 4.00, 4.00 for $B = 4, 8, 16, 32$ — it is below $|K|$ only because a lucky initial play occasionally hits the first type outright.

12 Discussion and Open Problems

Sections 9 and 10 locate the boundary of learnability for the unknown- K repeated security game. On one side, Theorem 6 shows that an unknown attacker set of bounded size costs only a factor $\sqrt{K_{\max}}$ as long as the defender is told, after each round, which type attacked. On the other side, Theorem 13 shows that

if the defender is told nothing but the attacked target, then every algorithm loses all but two of the T rounds — not because the adversary has many types, but because the defender cannot name the one type it faces. Theorem 16 quantifies this exactly, and in a currency that a security operations centre can price: on a universe of resolution M , and given *any* side channel that emits b bits over the whole horizon, the optimal regret is $\min\{T, \log_2 M - b\} \pm 3$. The attainable regret is the number of bits still needed to name the attacker; it is linear precisely when that number exceeds the horizon, and on an infinite universe no finite budget reduces it at all.

Between the two lies Problem 23, in which an oracle periodically reports the *set* of types encountered so far. The information it supplies is exactly a list of names; it says nothing about when or how often any of them attacked. Theorem 28 shows that this is enough: Algorithm 2 attains $O(nK_{\max}^{4/3}T^{2/3}\log^{1/3}(nK_{\max}) + K_{\max}B)$, the partial-information rate of Balcan et al. [2, Theorem 6.1] degraded by a factor $K_{\max}^{1/3}$ for the unknown catalogue and by an additive $K_{\max}B$ for the batching, and Theorem 32 shows the additive term is tight up to a factor $2B/(B-2)$, hence up to $2 + o(1)$ once B is large. Corollary 30 turns this into a deployment

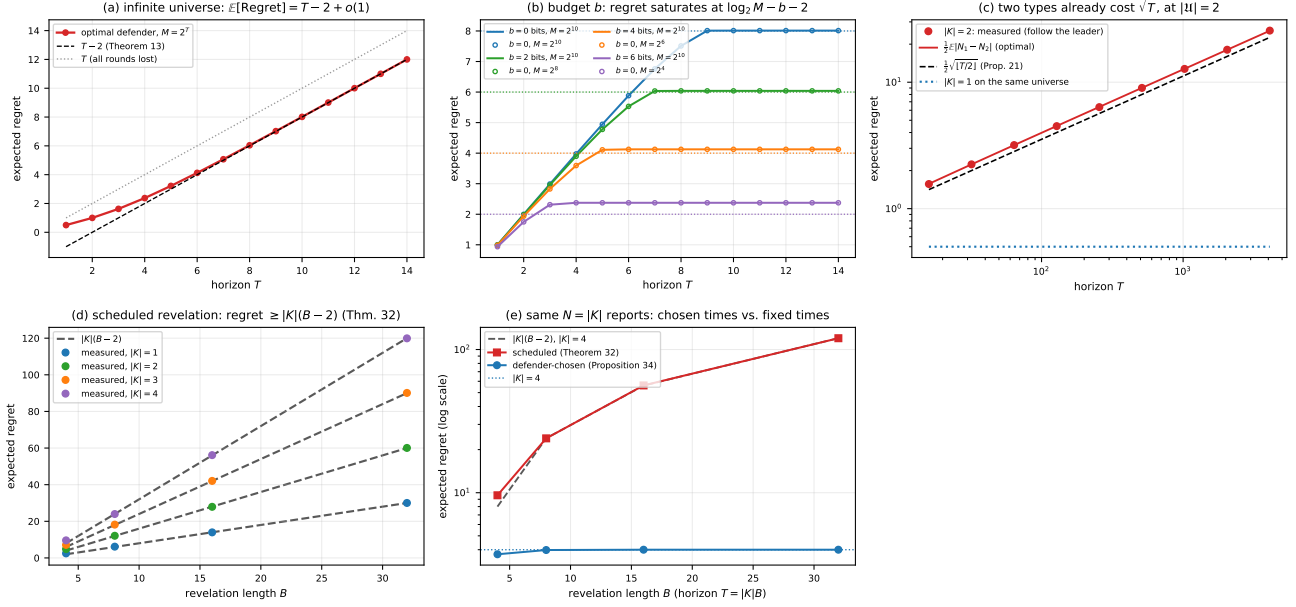


Figure 5: The gadget G_3 in simulation, against the optimal defender. (a) Theorem 13: with $|\mathcal{U}_M| = 2^T$ the regret is $T - 2 + o(1)$. (b) Theorem 16: with a budget of b bits on a universe of resolution $M = 2^{10}$ the regret saturates at $\log_2 M - b - 2$ (dotted lines), and a budget- b run on $\mathcal{U}_M$ (solid) coincides with a budget-0 run on $\mathcal{U}_{M/2^b}$ (hollow markers). (c) Proposition 21: two types on a two-element universe cost $\Theta(\sqrt{T})$, one type costs 0.5 at every horizon. (d) Theorem 32: on the nested instance the regret matches $|K|(B - 2)$ (dashed lines). (e) Proposition 34: the same $N = |K|$ reports cost $|K|(B - 2)$ on a fixed schedule and $|K|$ when the defender picks their times. Panels (c) and (e) have logarithmic vertical axes.

statement: the threat-intelligence report needs to arrive only $\tilde{O}(T^{1/3})$ times over a horizon of length T for the batching to be free — and, by the lower bound in the same corollary, not fewer.

Two views of the same picture from the online-learning literature are worth recording, because they say where these results sit. The query-budget statement of Theorem 16 is a security-game analogue of *label efficient prediction* [5], where a learner may see the loss vector on only a bounded number of rounds; the difference is that there the budget limits how *often* the learner may look, whereas Definition 15 limits how *much* it may be told in total, and consequently the rate here is governed by $\log_2 |\mathcal{U}|$ rather than by the number of experts. The additive $\Theta(K_{\max}B)$ of Theorems 28 and 32 is a delay phenomenon of the kind studied for general online learning under delayed feedback [11]; what is specific to the present setting is that the delay applies not to the loss but to the *identity* of the opponent, and that Theorem 13 makes the undelayed identity indispensable rather than merely helpful.

Two independent obstructions are thus isolated. Theorem 16 is about *volume*: too few bits, and the type cannot be named. Theorem 32 is about *latency*: even an unrestricted number of bits, delivered on a calendar of period B , still costs $\Theta(K_{\max}B)$, because the defender must play blind between the arrival of a type and the arrival of the report that names it. Proposition 34 shows the two are genuinely different resources: in G_3 , letting the defender choose *when* the same $K_{\max}$ reports arrive collapses the latency cost from $T - 2K_{\max}$ to $K_{\max}$.

Five questions are left open.

Is the exponent $2/3$ necessary here? Remark 29 observes that the exponent in Theorem 28 is inherited from Balcan et al. [2, Theorem 6.1] and is untouched by the unknown catalogue, so any improvement would improve their result. It is not known whether $T^{2/3}$ is optimal for partial-information feedback even with K known; the natural attack is to replace their explore-then-exploit estimator by importance-weighted estimates on the exploit rounds, which would require a version of Lemma 27 for a continuously updated estimator.

Is the factor $K_{\max}^{1/3}$ necessary? It arises from Hölder over the at most $K_{\max}$ epochs (Step 8), exactly as the $\sqrt{K_{\max}}$ of Theorem 6 arises from Cauchy–Schwarz (Step 9 of Section 6). Theorem 32 lower bounds only the additive term, and Proposition 21 lower bounds only the T -dependence. A lower bound of the form $\Omega(K_{\max}^c T^{2/3})$ for some $c > 0$, matching the multiplicative price of not knowing K , is open in both the full-information and the partial-information settings. The obvious attempt — running $K_{\max}/2$ disjoint copies of the two-type construction of Proposition 21, one per epoch — does not work, and the reason is instructive. Per-epoch lower bounds do not add, because the benchmark Equation (3) is a *single* fixed coverage vector: in G_3 with all the intervals disjoint, Lemma 11 says that no $\mathbf{p}$ hits more than one type at all, so the global benchmark is far weaker than the sum of the per-epoch benchmarks and the global regret is correspondingly smaller. A matching lower bound will have to make the epochs share a good comparator, as the nested family of Theorem 32 does, while still forcing fresh learning in each of them.

The complementary attack is on the upper bound: avoid

the restart, and both factors disappear with it, since with no epochs there is neither a Cauchy–Schwarz nor a Hölder step. Algorithms that track a *growing* expert set [17] do exactly that, and Section 2.2 already notes them as a drop-in replacement for the Polynomial-Weights sub-routine of Algorithm 1. What the proofs above show is that replacing the sub-routine is not by itself enough: Steps 4 and 5 of Section 6 also use the epoch structure, to pass from the epoch-local optimum to the global $\mathbf{p}^*$. Under partial information the difficulty is sharper still. The estimator of Lemma 36(ii) is defined relative to a fixed catalogue — its spanner Λ and the ambient set $\mathcal{W}$ both change when C grows — so the estimates produced before and after a discovery block are not estimates of comparable quantities, and cannot be handed to a single continuously running learner without a new argument.

Confounded search: $K_{\max} \geq 2$ on a finite universe. Theorem 16 pins the optimal regret on a resolution- M universe to $\min\{T, \log_2 M - b\} \pm 3$ when $K_{\max} = 1$, because a miss then identifies a half of the version space unambiguously. For $K_{\max} \geq 2$ the defender does not know which of its opponents produced the observation, so the same search is corrupted. Proposition 21 settles one half of the question — the answer must contain an $\Omega(\sqrt{T})$ term, already at $M = 2$ — and Remark 22 records the two upper bounds we can write down, $O(\sqrt{TK_{\max}(n^2 + K_{\max} \log n)})$ under full information and $O(T^{2/3}nM \log^{1/3}(nM))$ under partial information. Neither has any $\log M$ in it, and the lower bound has no M in it either, so the joint dependence on M and $K_{\max}$ is entirely open. Note that Theorem 13 does not settle it: it shows the $M \rightarrow \infty$ limit is $\Theta(T)$ for every $K_{\max} \geq 1$, but says nothing about the finite- M dependence for $K_{\max} \geq 2$.

One theorem for volume and latency together. Theorem 16 charges an oracle for its bits and ignores their timing; Theorem 32 charges for the timing and lets the bits be free. Neither subsumes the other, and a defender buying threat intelligence pays for both at once. The missing statement is a lower bound in terms of a pair (b, λ) — total budget and worst-case delay between a type’s first attack and the first message that names it — which should degrade to Theorem 16 at $\lambda = 0$ and to Theorem 32 at $b = \infty$. We expect the right form to be $\Omega(\min\{T, \log_2 M - b\} + K_{\max}\lambda)$, but the two arguments resist being run simultaneously: Lemma 12 conditions on a transcript that the latency argument needs to keep random. A related gap is the oracle whose messages are unbounded in length but weak in content — one that reports the utilities of a uniformly random element of D_ν , say. Such an oracle is not covered by Theorem 16, whose budget is a bound on message *length*, and the version-space construction of Theorem 16(ii) does not apply to it either.

Self-timed reports in a general game. Proposition 34 shows that in G_3 a defender that chooses its own report times pays $K_{\max}$ rather than $K_{\max}B$, because there a missed round is a certificate that the catalogue is stale. Remark 35 explains why the certificate is special to that gadget. Whether some other trigger — a statistical test on the estimator of Lemma 36(ii), say — recovers a comparable saving in a general repeated SSG, and whether the $\Theta(K_{\max}T/N)$ of Corollary 30

is the truth for self-timed queries, we leave open. This is the question with the most direct operational content: it asks whether it is better to subscribe to a periodic feed or to keep a budget of on-demand lookups.

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